WSCOM-YOLO: Directional-Reliability-Guided Local Spectral Communication for Aerial Small Object Detection

Xin Wang, Jiemin Wang, Mingchao Sun, Ce Song, Sheng Cai, Linlin Li, Jingwen Yan, and Dejiang Wang

***Abstract*—Aerial small-object detection is challenged by limited object pixels, dense layouts, and structured backgrounds that weaken boundaries and generate strong texture responses. We propose WSCOM-YOLO, which treats this task as weak local high-frequency signal recovery under structured interference. A signal-to-noise ratio (SNR)-inspired wavelet-SNR bottleneck (WSB) and wavelet multiresponse bridge (WMB) provide auxiliary low-/high-frequency premodulation; contour-selective fusion (CSF) combines anisotropic and contextual responses; and omnispectral communication (OSC) summarizes local spectral magnitudes from a real-input fast Fourier transform (FFT) as radial-angular tokens, processes them with a gated recurrent spectral scan (GRSS), and injects the reconstructed residual through a learned spatial gate. Across three seeds on VisDrone2019-DET validation, WSCOM-YOLO obtains a mean average precision (mAP) of 48.59% ± 0.23% at an intersection-over-union (IoU) threshold of 0.50 (mAP50) and 29.41% ± 0.23% averaged over IoU thresholds from 0.50 to 0.95 (mAP50–95), exceeding the same-framework YOLOv5s+P2 baseline by 2.96 and 2.11 percentage points. Three-seed model comparisons, including a billions of floating-point operations (GFLOPs)-matched control, are complemented by single-seed branch and dither controls, scale-specific evaluations, and visualizations. Dataset-specific results on DIOR and AI-TODv2 further indicate applicability to optical remote-sensing and tiny-object scenes. WSCOM-YOLO-Lite uses radial-only tokenization, achieving numerically similar three-seed mean accuracy on VisDrone with 41.78% lower forward latency than WSCOM-YOLO. The lite version is preferred for latency-sensitive applications; the full model should be considered when accuracy gains from target-dataset validation outweigh its higher forward latency.**

Received 20 July 2026; revised 17 September 2026 and 9 October 2026; accepted 14 October 2026. Date of publication 19 October 2026; date of current version 2 November 2026. This work was supported in part by the Plan for Scientific and Technological Development of Jilin Province under Grant SKL202402024, in part by the Natural Science Foundation of Guangdong Province under Grant 2024A1515011766, in part by the Scientific Research Start-Up Foundation of Guangzhou Institute of Science and Technology under Grant 2024KYQ004, and in part by Guangdong Key Discipline Research Capacity Enhancement Project under Grant 2025ZDJS089. *(Corresponding author: Dejiang Wang.)*

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Digital Object Identifier 10.1109/TGRS.2026.3744654

***Index Terms*—Aerial small-object detection, directional reliability, feature enhancement, frequency modeling, local spectral communication.**

I. Introduction

UNCREWED aerial vehicle (UAV) object detection supports traffic monitoring, urban governance, emergency response, and remote-sensing interpretation. Compared with natural-image detection, aerial detection involves smaller imaging scales, heavier occlusion, denser instance distributions, and more complex background textures. In VisDrone2019-DET [1], objects in categories such as pedestrians, bicycles, tricycles, awning-tricycles, and motors may occupy only a few pixels. Repeated downsampling and multiscale fusion can cause their contours, boundaries, and local directional cues to be overwhelmed by high-frequency background responses. Dense layouts, building reflections, and vegetation-background interference pose challenges for small-object detection in UAV-captured scenes.

The YOLO series provides an effective one-stage accuracy–speed tradeoff [2], [3]. A YOLOv5 framework adapted with P2 [4] extends the conventional P3/P4/P5 hierarchy with a stride-4 high-resolution branch, allowing shallow detail to participate directly in prediction. However, adding P2 alone does not resolve two issues: object boundaries and background textures remain entangled in the spatial domain, and weak small-object responses can be diluted by stronger activations during semantic fusion. We therefore view aerial small-object detection as weak high-frequency signal recovery under structured interference [5], [6], [7], [8], requiring both high-resolution features and reliability-aware selection of object evidence.

High-resolution heads, feature pyramids, attention, and context modules improve aerial detection [9], [10], [11], [12], [13], [14], yet mostly enhance generic spatial or semantic features. Increasing shallow resolution or fusion range can also amplify road markings, building edges, vegetation, and shadows. Because these responses can resemble small-object boundaries, resolution alone does not determine whether preserved detail is useful. This motivates investigating directional and spectral conditioning before cross-scale aggregation and prediction.

WSCOM-YOLO retains the YOLOv5 backbone-neck architecture consisting of C3 and SPPF blocks, where C3 denotes the cross-stage partial bottlenecks with three convolutions and SPPF denotes the spatial pyramid pooling-fast. The network builds four feature paths corresponding to P2–P5, and employs an Ultralytics anchor-free detection

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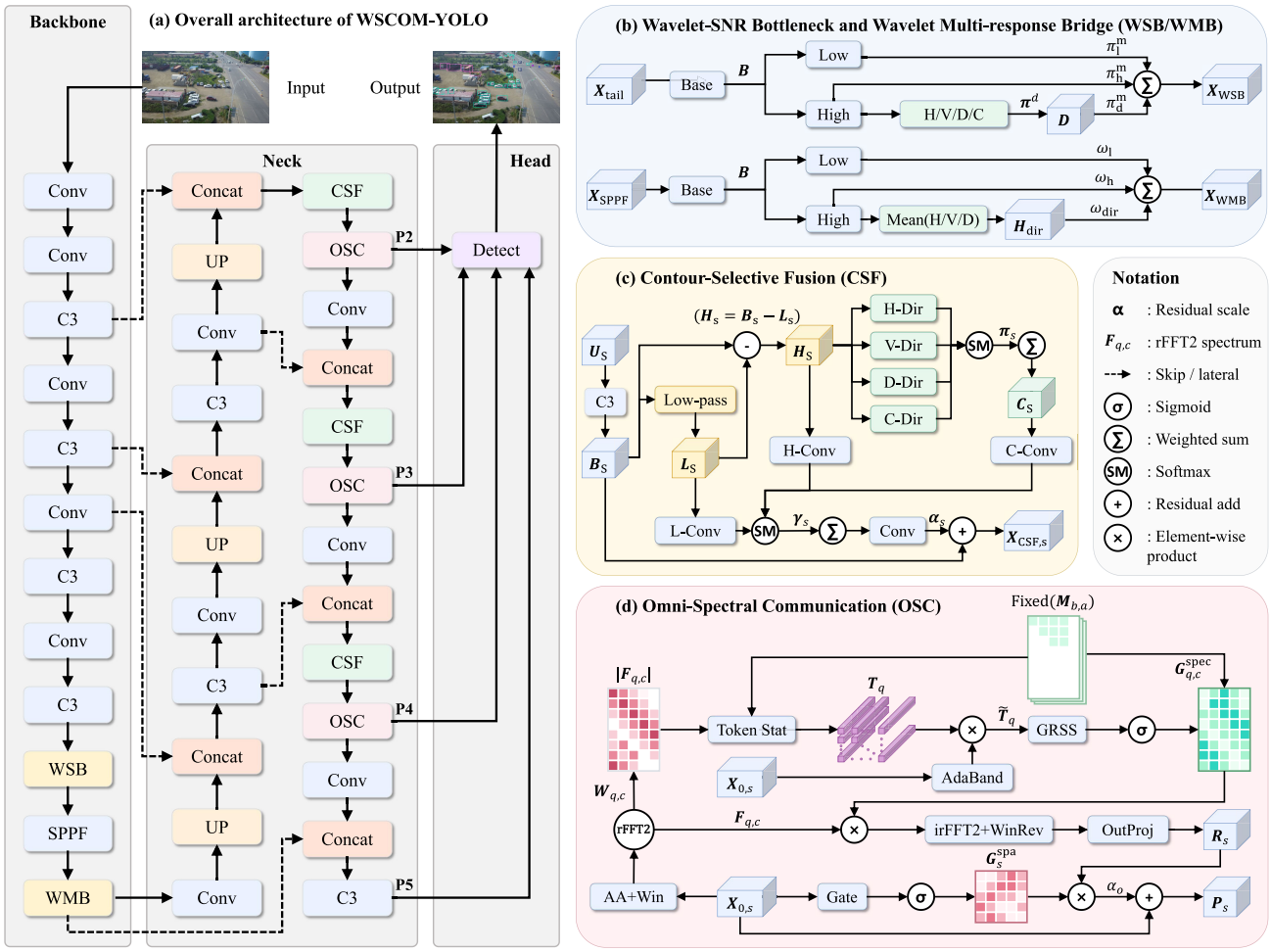


Fig. 1. WSCOM-YOLO architecture. (a) Four-scale backbone–neck and Detect head. (b) WSB/WMB premodulation. (c) CSF directional fusion. (d) OSC local spectral communication and gated injection. Operations omitted in (b) are defined in (12) and (16); repeated $X_{0,s}$ blocks in (d) same tensor.

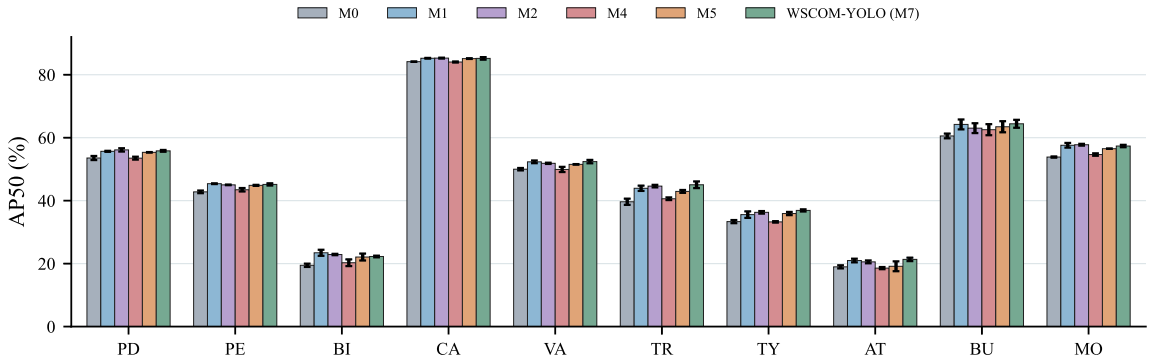


Fig. 2. Class-wise AP50 for M0, M1, M2, M4, M5, and WSCOM-YOLO (M7) on the VisDrone validation set using the fixed-version Ultralytics evaluator. Bars and error bars denote the mean and sample standard deviation over seeds 0, 1, and 2. Class abbreviations: PD, pedestrian; PE, people; BI, bicycle; CA, car; VA, van; TR, truck; TY, tricycle; AT, awning-tricycle; BU, bus; MO, motor.

head. The signal-to-noise ratio (SNR)-inspired wavelet-SNR bottleneck (WSB) and wavelet multiresponse bridge (WMB) provide auxiliary low-/high-frequency premodulation and learned response bridging at the backbone tail and feature pyramid network (FPN)/path aggregation network (PAN) entrance. The principal contour-selective fusion (CSF)–omnispectral communication (OSC) pathway then performs multibranch directional-response fusion, local radial-angular spectral-token communication, and spatially gated residual injection before prediction. This organization integrates front-end premodulation, spectral-residual generation, and spatial modulation into one end-to-end object detection framework.

The main contributions of this work are summarized as follows.

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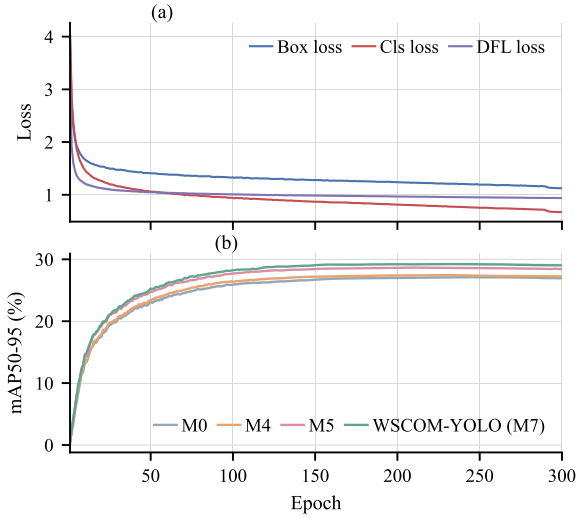


Fig. 3. Three-seed convergence over 300 epochs using the fixed-version Ultralytics validation evaluator. (a) Mean WSCOM-YOLO (M7) training losses with one-standard-deviation bands. (b) Validation mAP50–95 for M0, M4, M5, and M7. Here, M0, M4, M5, and M7 denote YOLOv5s+P2, no-CSF, no-OSC, and WSCOM-YOLO, respectively.

TABLE I
SCALE-SPECIFIC OSC CONFIGURATION OF WSCOM-YOLO

Level	Stride/C	w	$B/A/BA$	κ/λ_a	σ_d/N_d
P2	4/128	8	2/4/8	0.6/0.5	$10^{-4}/8$
P3	8/256	8	3/8/24	0.6/0.5	$10^{-4}/8$
P4	16/128	8	3/8/24	0.6/0.5	0/0

Note: Stride/C: input stride and equal input/output channels. w: window side length in feature-map pixels; B: number of radial bands; A: number of angular bins; BA: tokens per window. Shared settings: 3×3 average-pool anti-aliasing, radial-angular tokens, GRSS ($d_{\text{state}} = 16$, $\beta_{\text{ret}} = 0.7$, shared across all levels and experiments), AdaBand bottleneck reduction ratio 4, $\alpha_o = 0.25$, and 3×3 Conv–BN–SiLU block. Masks are fixed; final AdaBand/gate layers are zero-initialized. N_d: configured maximum number of dithered tail tokens per window. The efficiency-oriented WSCOM-YOLO-Lite configuration sets $A = 1$ at P2–P4, giving $BA = 2/3/3$ radial tokens; all other OSC and detector settings are unchanged.

1) We formulate aerial small-object detection as weak local high-frequency recovery under structured interference and organize WSCOM-YOLO into frequency premodulation, directional conditioning, and local spectral injection. This formulation motivates the architecture.
2) We design WSB and WMB as auxiliary front-end modules. WSB combines low-frequency context, high-frequency residuals, and a local high-to-low absolute-response ratio at the backbone tail, while WMB aggregates learned directional responses at the FPN/PAN entrance.
3) We develop a CSF-OSC prehead pathway. CSF fuses horizontal, vertical, local/diagonal, and dilated contextual responses; OSC communicates local radial-angular spectral tokens through a gated recurrent spectral scan (GRSS) and AdaBand and injects reconstructed residuals through a learned spatial gate.

II. RELATED WORK

A. Aerial/Remote-Sensing Benchmarks and Small Objects

Aerial benchmarks often contain small, densely distributed, and variably oriented objects against structured backgrounds. VisDrone2019-DET, UAVDT, DOTA, DIOR, NWPU VHR-10, and AI-TOD/NWD cover UAV-view, oriented-object, optical

TABLE II
COMPARISON ON THE VISDRONE VALIDATION SET (%)

Method	Precision	Recall	mAP50	mAP50-95
Research-group experiments under method-specific settings				
YOLOv5n‡ [4]	37.20	27.90	26.00	13.10
YOLOv5s‡ [4]	50.70	37.31	38.85	23.22
YOLOv8s‡ [26]	50.65	38.72	39.52	23.71
GBS-YOLOv5‡ [30]	47.70	36.40	36.80	20.60
Drone-YOLO‡ [14]	53.90	42.80	43.90	26.70
SuperYOLO-s‡ [13]	56.50	43.00	45.30	26.40
SOD-YOLOv8s‡ [32]	53.90	43.90	45.10	26.60
BRSTD‡ [33]	54.27	43.49	45.16	27.12
YOLOv5s+P2‡ [4]	53.51	43.51	44.93	27.21
YOLOv8s+P2‡ [26]	53.61	43.11	44.32	27.03
Fixed-version Ultralytics evaluation				
YOLO11s+P2 [26]	53.86±0.72	42.31±0.21	41.90±0.19	24.94±0.12
WSCOM-YOLO	**58.84±0.95**	**48.15±0.38**	**48.59±0.23**	**29.41±0.23**

Note: ‡ Results obtained in research-group experiments under method-specific training and evaluation settings; the citations identify the original methods. Unmarked rows use the fixed-version Ultralytics 8.4.41 evaluator and report the mean ± sample standard deviation across three independently trained seeds (0–2).

TABLE III
VISDRONE2019-DET TEST-DEV COMPARISON (1610 IMAGES; FIXED-VERSION ULTRALYTICS EVALUATION; %)

Method	Precision	Recall	mAP50	mAP50-95
YOLOv5s† [4]	44.97	33.99	31.18	17.56
YOLOv5s+P2† [4]	49.34	39.29	36.73	21.19
HIC-YOLOv5† [29]	48.49	39.35	36.45	20.94
YOLO11s+P2 [26]	45.94	37.29	33.94	19.34
WSCOM-YOLO	**51.94**	**41.24**	**38.80**	**22.69**

Note: † Ultralytics adaptations using an anchor-free head, DFL, and task-aligned assignment. HIC-YOLOv5† uses the validation-selected checkpoint from a 300-epoch seed-0 run.

TABLE IV
DIOR TEST COMPARISON UNDER FIXED-VERSION ULTRALYTICS EVALUATION (%)

Method	Precision	Recall	mAP50	mAP50-95
YOLOv5s† [4]	82.67	64.14	70.03	48.45
YOLOv5s+P2† [4]	83.30	63.59	70.58	49.09
HIC-YOLOv5† [29]	83.16	65.33	71.76	49.84
YOLO11s+P2 [26]	83.67	**66.69**	72.63	51.62
WSCOM-YOLO	**84.23**	66.50	**73.09**	**52.55**

Note: † Ultralytics adaptations using an anchor-free head, DFL, and task-aligned assignment. HIC-YOLOv5† is a local adaptation, not the original author implementation.

TABLE V
AI-TODV2 TEST COMPARISON UNDER FIXED-VERSION ULTRALYTICS EVALUATION (%)

Method	Precision	Recall	mAP50	mAP50-95
YOLOv5s† [4]	59.84	52.94	49.97	21.53
YOLOv5s+P2† [4]	63.06	53.92	53.21	23.62
HIC-YOLOv5† [29]	65.72	58.19	56.68	25.03
YOLO11s+P2 [26]	60.50	55.05	52.81	23.66
WSCOM-YOLO	**65.86**	**60.50**	**58.82**	**26.35**

Note: † Ultralytics adaptations using an anchor-free head, DFL, and task-aligned assignment. HIC-YOLOv5† is a local adaptation, not the original author implementation.

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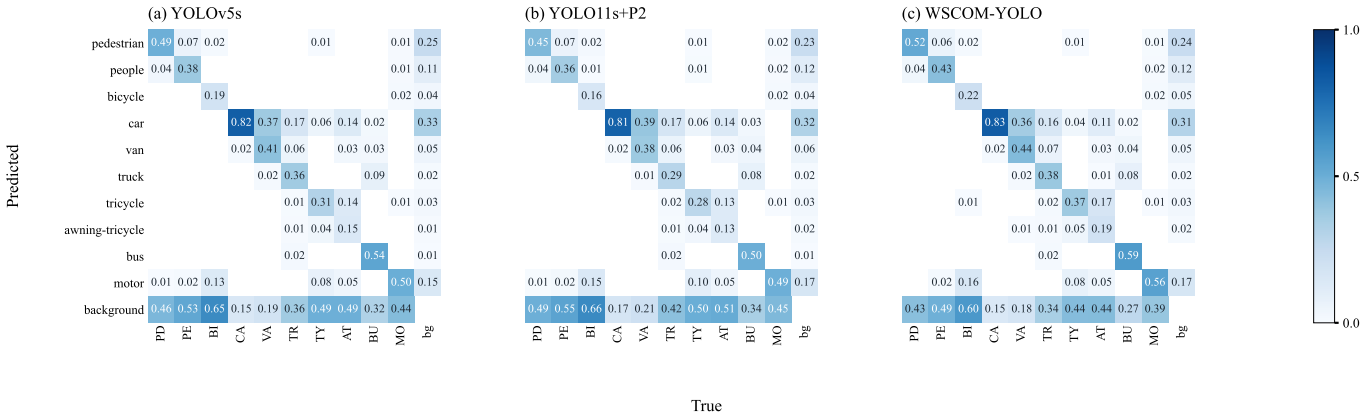


Fig. 4. Normalized VisDrone confusion matrices for M0, YOLO11s+P2, and WSCOM-YOLO from fixed-version Ultralytics validation predictions, using score > 0.25, IoU > 0.45, and a shared [0, 1] scale. Values below 0.005 remain color-encoded but unannotated. Class abbreviations: PD, pedestrian; PE, people; BI, bicycle; CA, car; VA, van; TR, truck; TY, tricycle; AT, awning-tricycle; BU, bus; MO, motor. The label bg denotes background. Each matrix uses one fixed checkpoint. (a) YOLOv5s. (b) YOLO11s+P2. (c) WSCOM-YOLO.

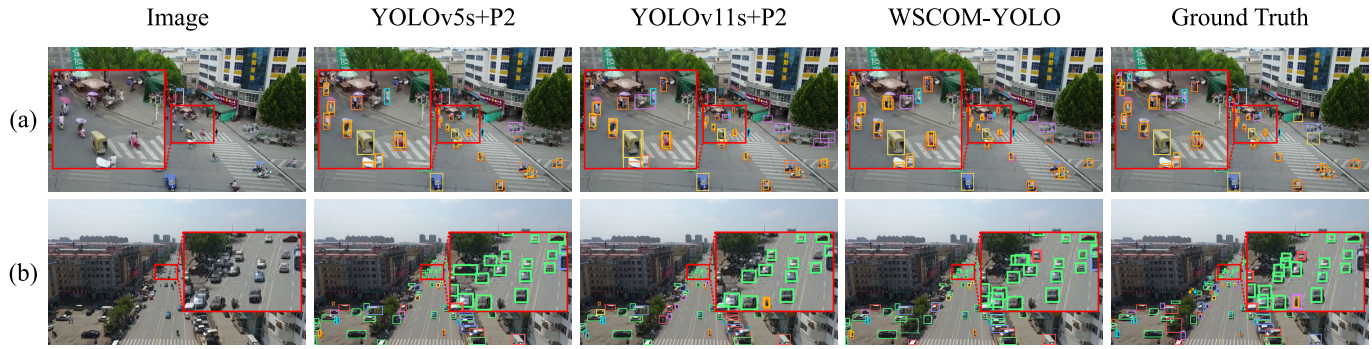


Fig. 5. Detection examples on the VisDrone validation set (a) pedestrians at an intersection and (b) vehicles along an urban road. Red boxes mark regions shown in enlarged insets.

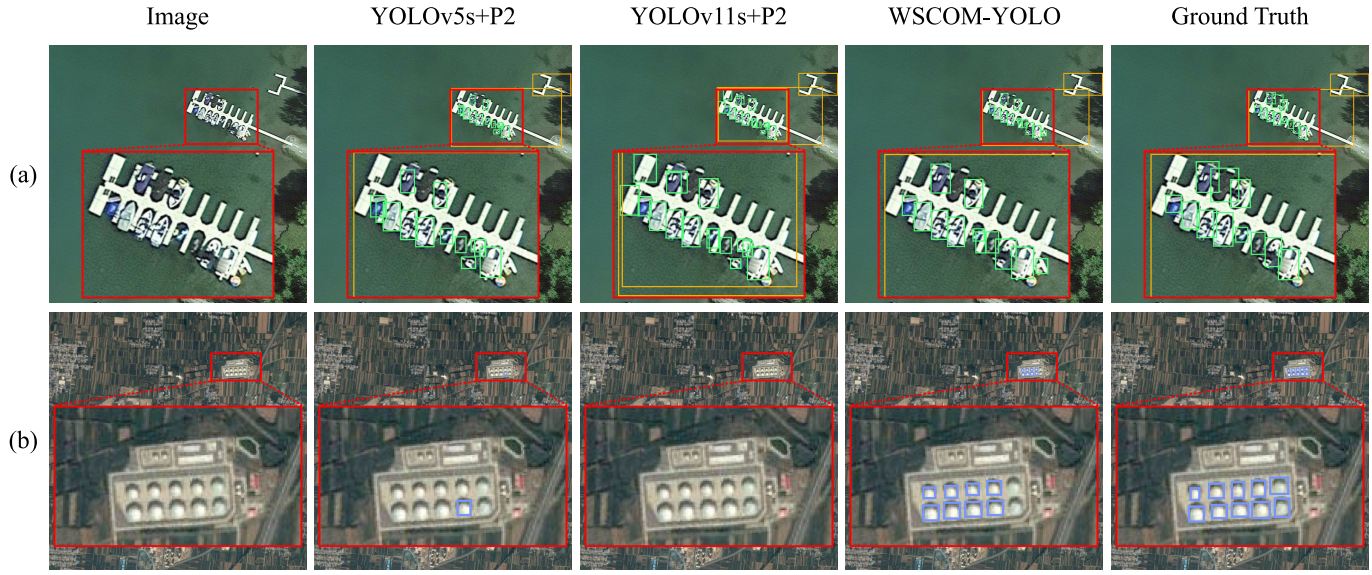


Fig. 6. DIOR validation detection examples (a) harbor and ships and (b) storage tanks. Red boxes mark regions shown in enlarged insets.

remote-sensing, and tiny-object settings [1], [15], [16], [17], [18], [19]; surveys identify scale, orientation, clutter, and weak representation as persistent challenges [20], [21]. These challenges call for feature representations that preserve small-object details while suppressing background interference.

B. Multiscale Detectors and UAV Small Object Methods

Region-proposal learning in Faster R-CNN [22] and one-stage prediction in YOLO [2], [3] represent two widely used detection frameworks. In these frameworks, FPN and PANet improve cross-scale feature propagation [9], [10], while

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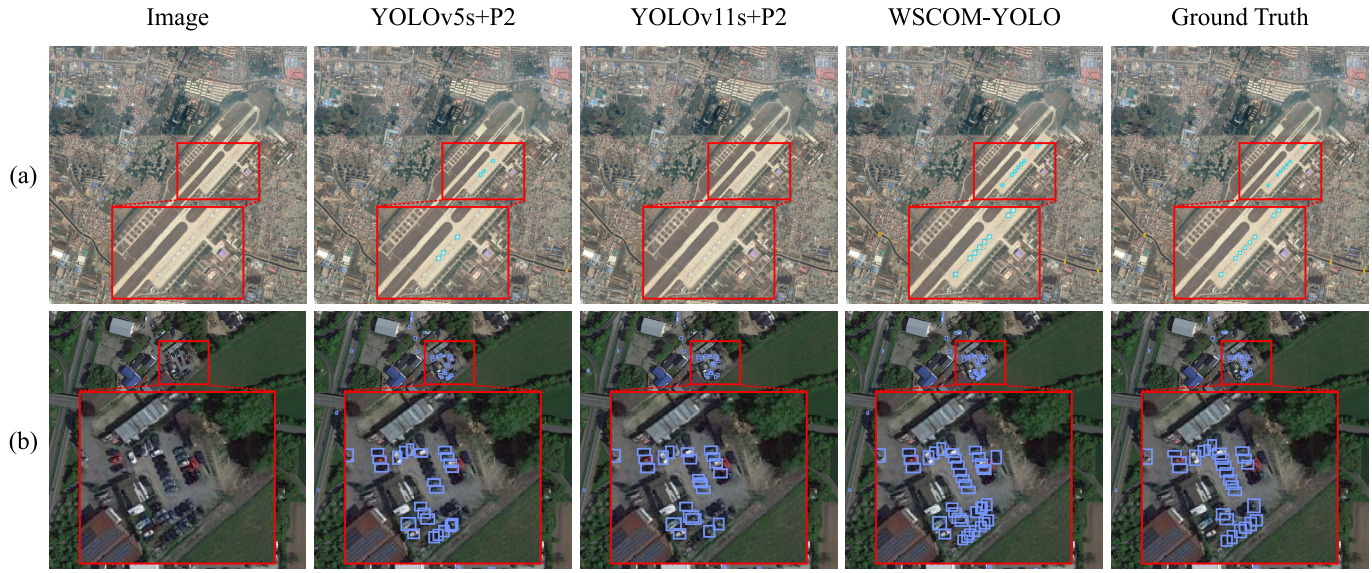


Fig. 7. AI-TODv2 test detection examples (a) tiny airplanes and (b) clustered vehicles. Red boxes mark regions shown in enlarged insets.

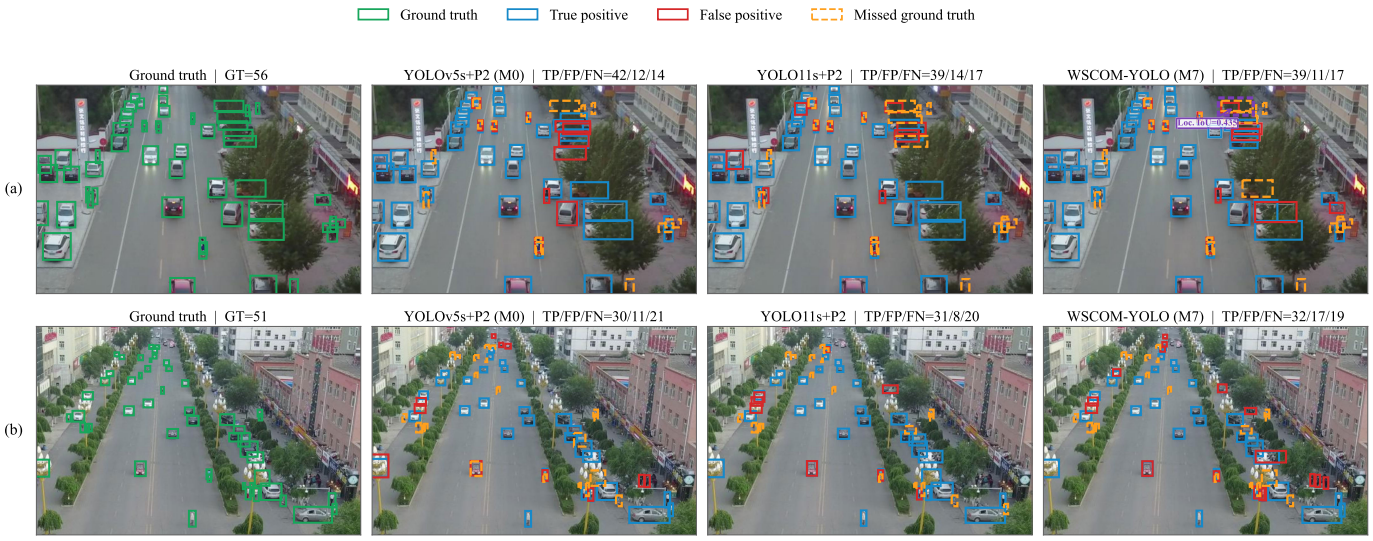


Fig. 8. Representative VisDrone failure modes (a) weak-evidence and localization failures under occlusion and scale variation and (b) false positives amid structured background texture. Green/blue/red/orange dashed boxes denote ground truth/true positives/false positives/misses; the purple callout marks a class-correct localization miss (IoU = 0.435). Crop counts use score $\geq$ 0.25 and class-aware one-to-one matching at IoU $\geq$ 0.5.

TABLE VI

COMMON-EVALUATOR COMPARISON ON THE VISDRONE VALIDATION SET USING PYCOCOTOOLS COCOEVAL WITH A MAXIMUM OF 300 DETECTIONS PER IMAGE (maxDets = 300). THIS COMPARISON USES FIXED CHECKPOINTS; WSCOM-YOLO USES ITS SEED-0 CHECKPOINT. ACCURACY IS REPORTED IN %

Method	mAP50	mAP50-95
SMWG-DETR (official 36-epoch recipe) [60]	28.54	14.99
DCP-Net [61]	39.58	23.26
HIC-YOLOv5 [29]	40.65	23.82
WSCOM-YOLO	**48.91**	**29.82**

RetinaNet addresses the foreground–background imbalance in dense prediction [23]. EfficientDet and MobileNetV2 improve computational efficiency through architecture scaling and efficient building blocks [24], [25]. For aerial small objects, these developments provide a hierarchical feature backbone that can be extended with a high-resolution P2 prediction path [4], [26].

ClusDet and YOLC exploit spatial clustering [12], [27], whereas QueryDet employs sparse queries for high-resolution detection [28]. SuperYOLO, Drone-YOLO, HIC-YOLOv5, GBS-YOLOv5, HSP-YOLOv8, and SOD-YOLOv8 improve UAV detectors through multiscale fusion, attention, efficient design, or small-object heads [13], [14], [29], [30], [31], [32]; BRSTD studies the accuracy–complexity trade-off with antagonistic receptive fields and feedback suppression [33].

Recent methods address assignment, denoising, and contextual aggregation. KLDet reformulates tiny-object matching and localization quality through Kullback–Leibler divergence [34]. SWIN-TOD combines neighboring enhancement with the smooth Wasserstein distance [35]; DNTR suppresses feature-pyramid fusion noise via DeNoising FPN and enhances the

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TABLE VII
SAME-FRAMEWORK COMPLEXITY AND BATCH-1 FORWARD-ONLY INFERENCE SPEED

Method	Params (M)	GFLOPs	Latency (ms)	FPS
M0 (YOLOv5s+P2)	11.642	62.189	4.645 ± 0.053	215.29
M1 (no WSB)	17.664	148.873	22.359 ± 0.063	44.72
M2 (no WMB)	17.538	148.773	22.321 ± 0.113	44.80
M4 (no CSF)	15.039	86.632	20.611 ± 0.105	48.52
M5 (no OSC)	17.206	126.803	8.422 ± 0.040	118.74
GFLOPs-matched M0	18.995	149.550	7.301 ± 0.032	136.97
WSCOM-YOLO-Lite	18.602	145.021	13.353 ± 0.046	74.89
WSCOM-YOLO	18.602	149.633	22.934 ± 0.076	43.60

Note: WSCOM-YOLO corresponds to the M7 configuration used in the ablation studies; FPS denotes frames per second. FP32 network-forward latency at 640×640, batch 1, on one RTX 4090 with TensorFloat-32 (TF32) disabled; 50 warmup iterations and 300 CUDA-synchronized repetitions per round. All rows report the mean ± sample standard deviation of three round medians. Preprocessing, host-device transfer, and NMS are excluded.

TABLE VIII
CONDITIONAL MODULE COST FROM FUSED INFERENCE GRAPHS

Component	ΔParams (M)	ΔGFLOPs	ΔLatency (ms)
WSB	0.938	0.76	0.575 ± 0.115
WMB	1.064	0.86	0.613 ± 0.107
CSF (three scales)	3.562	63.00	2.322 ± 0.097
OSC (three scales)	1.396	22.83	14.512 ± 0.057

Note: M7 minus the corresponding fused module-removal graph at 640×640, computed round-wise from the unified profiling session and reported as mean ± sample standard deviation. The differences are conditional and non-additive.

representation of tiny objects with Transformer R-CNN [36]; and CAF^{2}ENet integrates cross-attention fusion, regression-guided refocusing, and feature alignment [37]. Global–local fusion, CFENet, S^{2}FCNet, and you only look denseness (YOLD) further model global and local context, spatial semantics, and object density [38], [39], [40], [41]. In contrast to these methods, we propose WSCOM-YOLO, which focuses on directional reliability and local radial-angular dependencies within a four-scale detector.

C. Directional Responses and Context Modeling in Remote Sensing

Orientation variation has motivated the development of rotation-invariant features, transformed RoIs, feature alignment, equivariance, and oriented proposals in rotation-invariant convolutional neural network (RICNN), region of interest (ROI) Transformer, SCRDet, R3Det, S2A-Net, ReDet, and Oriented R-CNN [42], [43], [44], [45], [46], [47], [48]; CAD-Net emphasizes contextual discrimination [49]. These studies show that directional structure is useful beyond oriented-box regression. Even with horizontal boxes, tiny objects may retain only short edges, corners, or contours, while road markings, roof boundaries, vegetation, and shadows can produce similar responses. Drawing on these observations, our proposed CSF selects anisotropic, local, and contextual candidates, upon which our OSC further models local radial-angular dependencies.

D. Frequency Modeling, Wavelet, and Contourlet Representations

Wavelets and Laplacian pyramids separate low-frequency structure from residual detail [5], [6], [7]. Soft thresholding and wavelet-domain statistics show that high-frequency components contain both useful edges and noise, and therefore require reliability selection [50], [51]. Contourlet and steerable representations further model the directional structure of 2-D edges and contours [52], [53], [54]. Classical edge, scale-space, diffusion, and local time-frequency analyses provide related foundations through gradients, zero crossings, denoising, scale selection, and localized spectra [55], [56], [57], [58], [59].

Recent tiny-object detectors introduce explicit frequency representations: SMWG-detection transformer (DETR) combines Fourier modulation with wavelet-guided fusion [60], DCP-Net learns detail-context perception under spatial-frequency guidance [61], and amplitude-phase recalibration treats spectral amplitude and phase separately [62]. By contrast, WSCOM-YOLO forms a staged pathway that proceeds from low- and high-frequency premodulation, through directional fusion, local radial-angular token communication, to spatially gated residual injection.

The fast Fourier transform (FFT) enables efficient local spectra [63]; Welch and Harris provide the foundation for band statistics and leakage-aware window analysis [64], [65]; and Shannon/Neyman–Pearson theory underpins reliable response discrimination under uncertainty [8], [66]. Local high-frequency responses can arise from both object details and background textures. Throughout this article, "spectral" refers to the frequency-domain representation of learned feature maps; the detector input remains a standard RGB image, and no additional multispectral or hyperspectral sensor bands are used. Motivated by this observation, we combine directional responses and local spectral statistics to adaptively modulate features.

E. Attention, Sequential Modeling, and Local Spectral Tokens

Fourier-domain token mixing provides a closely related precedent. FNet replaces self-attention with a parameter-free Fourier transform to mix a complete token sequence [67]. GFNet applies a 2-D discrete Fourier transform, a learnable global filter, and a 2-D inverse Fourier transform to capture long-range spatial dependencies [68]. FcaNet instead selects discrete-cosine-transform components to form compact channel-wise descriptors for channel recalibration [69]. Most closely related to adaptive filtering, the adaptive frequency filtering (AFF) token mixer in AFFNet predicts semantic-adaptive frequency filters and uses them as efficient global token mixers [70]. These methods show that fixed, learned, and content-adaptive spectral operators can replace global interaction or improve channel representation.

OSC shares frequency-domain modulation with these methods but differs in its modeling unit, spectral parameterization, and output pathway. Unlike AFFNet, which predicts content-adaptive global frequency filters, OSC partitions feature maps into nonoverlapping local windows, summarizes each spectrum with radial-angular masks, and uses GRSS to communicate across the resulting short radial-angular token sequence. The mixed tokens modulate the local spectra and reconstruct spatial residuals that, unlike FcaNet's channel-recalibration descriptors, are injected into the P2–P4 detection branches

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TABLE IX
SINGLE-CHECKPOINT M0–M7 ABLATION ON THE VISDRONE VALIDATION SET (FIXED-VERSION ULTRALYTICS EVALUATION; %)

ID	WSB	WMB	CSF	OSC	PD	PE	BI	CA	VA	TR	TY	AT	BU	MO	mAP50	mAP50-95
M0	–	–	–	–	52.80	42.47	19.97	84.13	49.95	39.29	32.85	18.63	59.98	54.04	45.41	27.21
M1	–	✓	✓	✓	56.22	45.34	22.47	85.41	51.99	44.65	34.47	21.38	63.48	57.30	48.27	29.30
M2	✓	–	✓	✓	56.02	44.96	23.14	85.32	51.51	44.22	36.13	20.87	64.33	57.88	48.44	29.20
M3	–	–	✓	✓	56.16	46.19	24.18	84.74	51.94	44.05	35.43	22.69	63.28	57.70	48.64	29.52
M4	✓	✓	–	✓	53.55	43.22	19.63	83.98	50.07	41.09	33.61	17.48	62.23	54.43	45.93	27.51
M5	✓	✓	✓	–	55.31	45.08	21.18	85.00	51.44	42.95	36.44	19.64	62.96	56.45	47.65	28.84
M6	✓	✓	–	–	53.68	43.24	19.61	84.03	48.94	38.20	33.44	19.06	60.03	54.21	45.44	27.12
M7	✓	✓	✓	✓	56.94	45.87	23.52	85.05	52.18	43.60	37.90	21.95	65.30	58.13	**49.04**	**29.67**

Note: PD–pedestrian, PE–people, BI–bicycle, CA–car, VA–van, TR–truck, TY–tricycle, AT–awning-tricycle, BU–bus, MO–motor. Per-class and overall metrics come from the same evaluation. These checkpoints differ from those used for the three-seed experiments reported in Section IV-D2 and the seed-0 controls reported in Section IV-D3.

through a separate learned spatial gate. The corresponding accuracy–efficiency results are reported in Section IV.

Squeeze-and-excitation (SE), Convolutional Block Attention Module (CBAM), and efficient channel attention (ECA) show that channel and spatial recalibration can improve feature selection [11], [71], [72]. Deformable convolution and nonlocal networks model adaptive or global spatial context [73], [74]. Transformer and DETR promote attention-based end-to-end detection, while Deformable DETR reduces the cost of global interaction through sparse sampling [75], [76], [77]. Recurrent links and gating mechanisms provide dynamic memory and regulated information transmission, motivating short-sequence propagation and input selection in GRSS [78], [79].

Serializing an entire aerial feature map can mix weak objects with distant structured texture. OSC therefore applies GRSS only to radial-angular tokens within each window, combining 1-D depthwise convolution, recurrence, and token-conditioned gating. A learned spatial gate then modulates reconstructed residuals under the end-to-end detection loss.

III. METHOD

A. Signal Modeling and Directional-Spectral Enhancement Pathway

We represent an aerial detection feature at pyramid level s as

$$\boldsymbol{X}_s = \boldsymbol{X}_s^{\mathrm{obj}} + \boldsymbol{X}_s^{\mathrm{bg}} + \boldsymbol{X}_s^{\mathrm{intf}} \tag{1}$$

where $\boldsymbol{X}_s \in \mathbb{R}^{C_s \times H_s \times W_s}$ is the learned feature tensor, and C_s, H_s, and W_s are its channel, height, and width. The semantic superscripts obj, bg, and intf identify the sparse local object response, slowly varying background response, and structured interference including clutter, aliasing, and pseudo high-frequency texture, respectively. This is a conceptual decomposition motivated by multiresolution signal analysis and weak-signal detection [5], [6], [7], [66]. Let $\mathcal{D}_s(\cdot)$ map a feature from level s to level $s+1$, and let $\mathcal{P}_{\mathrm{h}}(\cdot)$ extract its high-frequency component. Under the low-pass/downsampling interpretation of hierarchical feature extraction

$$\boldsymbol{X}_{s+1}^{\mathrm{obj}} = \mathcal{D}_s\left(\boldsymbol{X}_s^{\mathrm{obj}}\right), \quad \left\|\mathcal{P}_{\mathrm{h}}\left(\boldsymbol{X}_{s+1}^{\mathrm{obj}}\right)\right\|_2 \le \left\|\mathcal{P}_{\mathrm{h}}\left(\boldsymbol{X}_s^{\mathrm{obj}}\right)\right\|_2 \tag{2}$$

where $\|\cdot\|_2$ is the Euclidean norm; the inequality states a nonexpansive low-pass/downsampling assumption on the high-frequency component of the object.

The implemented modules use a measurable local response ratio ρ as a reliability statistic. For any spatial position $\boldsymbol{p} = (x, y)$, we define

$$\rho(\boldsymbol{p}) = \frac{\eta(\boldsymbol{H})(\boldsymbol{p})}{\eta(\boldsymbol{L})(\boldsymbol{p}) + \varepsilon}, \quad \eta(\boldsymbol{Z}) = \frac{1}{C'}\sum_{c=1}^{C'} |\boldsymbol{Z}_c| \tag{3}$$

where $\boldsymbol{L}, \boldsymbol{H} \in \mathbb{R}^{C \times H \times W}$ are the local low- and high-frequency responses; $\boldsymbol{Z}_c$ is the cth channel of an arbitrary feature tensor $\boldsymbol{Z}$, C' is the channel count of $\boldsymbol{Z}$; and $\eta(\cdot)$ computes a channel-mean absolute-response map; $\varepsilon = 10^{-6}$ prevents division by zero, with the same ε adopted for all module-level reliability ratios. This SNR-inspired formulation defines a high-to-low absolute-response statistic that conditions learned fusion weights.

Let $\mathcal{F}_{\mathrm{tail}}(\cdot)$ denote the backbone mapping before WSB. The WSB and WMB perform auxiliary low-/high-frequency premodulation, which can be expressed as

$$\boldsymbol{X}_{\mathrm{wsb}} = \Psi_{\mathrm{wsb}}(\mathcal{F}_{\mathrm{tail}}(\boldsymbol{I})), \quad \boldsymbol{X}_{\mathrm{wmb}} = \Psi_{\mathrm{wmb}}(\chi(\boldsymbol{X}_{\mathrm{wsb}})) \tag{4}$$

where $\boldsymbol{I}$ is the input image; $\boldsymbol{X}_{\mathrm{wsb}}$ and $\boldsymbol{X}_{\mathrm{wmb}}$ are the WSB and WMB outputs; $\Psi_{\mathrm{wsb}}(\cdot)$ and $\Psi_{\mathrm{wmb}}(\cdot)$ denote the corresponding module mappings; and $\chi(\cdot)$ is the SPPF operation.

In the PAN bottom-up path, $\mathcal{N}_2(\cdot)$ constructs the preenhancement P2 branch, $\mathcal{N}_s(\cdot,\cdot)$ for $s \in \{3,4\}$ combines neck features derived from $\boldsymbol{X}_{\mathrm{wmb}}$ with the downsampled preceding enhanced branch, and $\mathcal{N}_5(\cdot,\cdot)$ constructs P5 with its native C3 mapping. The implemented pathway is expressed as

$$\boldsymbol{P}_s = \begin{cases} \Psi_{\mathrm{osc}}^2\left(\Psi_{\mathrm{csf}}^2\left(\mathcal{N}_2\left(\boldsymbol{X}_{\mathrm{wmb}}\right)\right)\right), & s = 2 \\ \Psi_{\mathrm{osc}}^s\left(\Psi_{\mathrm{csf}}^s\left(\mathcal{N}_s\left(\boldsymbol{X}_{\mathrm{wmb}}, \boldsymbol{P}_{s-1}\right)\right)\right), & s = 3,4 \\ \mathcal{N}_5\left(\boldsymbol{X}_{\mathrm{wmb}}, \boldsymbol{P}_4\right), & s = 5 \end{cases} \tag{5}$$

where $\Psi_{\mathrm{csf}}^s(\cdot)$ and $\Psi_{\mathrm{osc}}^s(\cdot)$ denote the module mappings for CSF and OSC, respectively; and $\boldsymbol{P}_s$ denotes the feature tensor at level s. Equations (4) and (5) preserve the implemented bottom-up dependence $\boldsymbol{P}_2 \to \boldsymbol{P}_3 \to \boldsymbol{P}_4 \to \boldsymbol{P}_5$, while the CSF-OSC pathway is applied only to the P2, P3, and P4 branches.

B. Overall Architecture of WSCOM-YOLO

Given an RGB input image $\boldsymbol{I} \in \mathbb{R}^{3 \times H_{\mathrm{in}} \times W_{\mathrm{in}}}$ of height H_{in} and width W_{in}, WSCOM-YOLO retains a YOLOv5 C3/SPPF backbone–neck and supplies four feature levels $\{\boldsymbol{P}_2, \boldsymbol{P}_3, \boldsymbol{P}_4, \boldsymbol{P}_5\}$ to the anchor-free Ultralytics detect head. As shown in Fig. 1(a), WSB (Section III-C) is placed

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at the backbone tail, WMB (Section III-D) between SPPF and FPN/PAN, and CSF (Section III-E) followed by OSC (Section III-F) on the P2/P3/P4 branches. The stride-4 P2 branch preserves high-resolution detail, whereas P5 follows the native spatial path because of its lower spatial resolution and stronger semantic response.

C. Wavelet-SNR Bottleneck

The WSB performs low-/high-frequency reliability premodulation at the backbone tail. For its input $\boldsymbol{X}_{\text{tail}} = \mathcal{F}_{\text{tail}}(\boldsymbol{I})$, we first construct a bottleneck representation via

$$\boldsymbol{B} = \varphi_{3\times3}\left(\varphi_{1\times1}\left(\boldsymbol{X}_{\text{tail}}\right)\right) \tag{6}$$

where $\boldsymbol{B}$ denotes the base feature and $\varphi_{a\times b}(\cdot)$ denotes an independent Conv–BN–SiLU block with kernel size $a \times b$. To decompose $\boldsymbol{B}$ into its low- and high-frequency components, we apply a stride-1 3×3 average pooling operation, denoted as $\mathcal{A}_{3\times3}(\cdot)$, which gives

$$\boldsymbol{L} = \mathcal{A}_{3\times3}\left(\boldsymbol{B}\right), \qquad \boldsymbol{H} = \boldsymbol{B} - \boldsymbol{L} \tag{7}$$

where $\boldsymbol{L}$ and $\boldsymbol{H}$ are low-frequency context and high-frequency residual. This decomposition is wavelet-inspired, not a discrete wavelet transform [5], [6], [7].

To extract multiorientation high-frequency candidate responses, we instantiate four trainable depthwise Conv–BN–SiLU blocks operating on $\boldsymbol{H}$

$$\begin{aligned} \boldsymbol{D}_{\text{h}} &= \psi_{1\times5}\left(\boldsymbol{H}\right), & \boldsymbol{D}_{\text{v}} &= \psi_{5\times1}\left(\boldsymbol{H}\right) \\ \boldsymbol{D}_{\text{d}} &= \psi_{3\times3}\left(\boldsymbol{H}\right), & \boldsymbol{D}_{\text{c}} &= \psi^{2}_{3\times3}\left(\boldsymbol{H}\right) \end{aligned} \tag{8}$$

where $\psi_{1\times5}(\cdot)$, $\psi_{5\times1}(\cdot)$, $\psi_{3\times3}(\cdot)$, $\psi^{2}_{3\times3}(\cdot)$ denote the independent depthwise convolution branches with kernel sizes 1×5, 5×1, 3×3, and dilation-2 3×3, respectively, capturing horizontal, vertical, local/diagonal, and dilated-contour cues.

A gate $\mathcal{G}_{1\times1}(\cdot)$, consisting of a 1×1 Conv–SiLU–Conv block, takes the channel-wise concatenation of seven response statistics to produce directional fusion weights

$$\begin{aligned} \boldsymbol{\pi}^{\text{d}} = \text{softmax}\Big(\mathcal{G}_{1\times1}\big[&\eta\left(\boldsymbol{B}\right), \eta\left(\boldsymbol{L}\right), \eta\left(\boldsymbol{H}\right) \\ &\eta\left(\boldsymbol{D}_{\text{h}}\right), \eta\left(\boldsymbol{D}_{\text{v}}\right), \eta\left(\boldsymbol{D}_{\text{d}}\right), \eta\left(\boldsymbol{D}_{\text{c}}\right)\big]\Big) \end{aligned} \tag{9}$$

where $[\cdot]$ is channel concatenation; the gate follows a $7 \rightarrow 16 \rightarrow 4$ channel configuration, and its final layer weights and biases are zero-initialized. The resulting softmax weights $\boldsymbol{\pi}^{\text{d}} = \{\pi^{\text{d}}_{\text{h}}, \pi^{\text{d}}_{\text{v}}, \pi^{\text{d}}_{\text{d}}, \pi^{\text{d}}_{\text{c}}\}$ are applied per spatial position to fuse the four directional candidates

$$\boldsymbol{D} = \sum_{i\in\{\text{h,v,d,c}\}} \pi^{\text{d}}_{i}\boldsymbol{D}_{i}. \tag{10}$$

We next compute a local reliability map $\boldsymbol{\rho}_1$ following the SNR-inspired formulation in (3). The low- and high-frequency components are then independently refined via $\boldsymbol{L}' = \varphi_{1\times1}(\boldsymbol{L})$ and $\boldsymbol{H}' = \varphi_{1\times1}(\boldsymbol{H})$. A second gate then predicts the aggregated weights

$$\begin{aligned} \boldsymbol{\pi}^{\text{m}} = \text{softmax}\Big(\mathcal{G}_{1\times1}\big[&\eta\left(\boldsymbol{B}\right), \eta\left(\boldsymbol{L}'\right), \\ &\eta\left(\boldsymbol{H}'\right), \text{clip}\left(\boldsymbol{\rho}_1, 0, 8\right)\big]\Big) \end{aligned} \tag{11}$$

where $\text{clip}(\cdot)$ truncates $\boldsymbol{\rho}_1$ to $[0, 8]$, and $\boldsymbol{\pi}^{\text{m}} = \{\pi^{\text{m}}_{\text{l}}, \pi^{\text{m}}_{\text{h}}, \pi^{\text{m}}_{\text{d}}\}$ weights low, high, and directional candidates. This gate follows a $4 \rightarrow 16 \rightarrow 3$ channel configuration, with SiLU activation in the hidden layer and zero-initialized final layer weights and biases.

The final WSB output is obtained via a residual aggregation

$$\begin{aligned} \boldsymbol{X}_{\text{wsb}} = \boldsymbol{X}_{\text{tail}} + \boldsymbol{B} \\ + \alpha_1\varphi_{1\times1}\Big(\pi^{\text{m}}_{\text{l}}\boldsymbol{L}' + \pi^{\text{m}}_{\text{h}}\boldsymbol{H}' + \pi^{\text{m}}_{\text{d}}\boldsymbol{D}\Big) \end{aligned} \tag{12}$$

where residual scale α_1 is initialized to 0.1. Note that since each $\boldsymbol{D}_i$ is already refined, only $\boldsymbol{L}$ and $\boldsymbol{H}$ receive branch refinements prior to the final projection $\varphi_{1\times1}(\cdot)$. The $\boldsymbol{X}_{\text{tail}}$ shortcut is used only when dimensions match.

D. Wavelet Multiresponse Bridge

WMB integrates low-frequency context and directional high-frequency responses before the FPN/PAN. For $\boldsymbol{X}_{\text{sppf}} = \chi(\boldsymbol{X}_{\text{wsb}})$, we first extract the base feature $\boldsymbol{B}$ and subsequently perform the same low-/high-frequency decomposition adopted in WSB

$$\boldsymbol{B} = \varphi_{1\times1}\left(\boldsymbol{X}_{\text{sppf}}\right), \quad \boldsymbol{L} = \mathcal{A}_{3\times3}\left(\boldsymbol{B}\right), \quad \boldsymbol{H} = \boldsymbol{B} - \boldsymbol{L}. \tag{13}$$

To enrich the high-frequency representation, we apply horizontal, vertical, and local/diagonal trainable depthwise branches: $\boldsymbol{H}_{\text{h}} = \psi_{1\times3}(\boldsymbol{H})$, $\boldsymbol{H}_{\text{v}} = \psi_{3\times1}(\boldsymbol{H})$, and $\boldsymbol{H}_{\text{d}} = \psi_{3\times3}(\boldsymbol{H})$. Their outputs are averaged to obtain an aggregated directional response

$$\boldsymbol{H}_{\text{dir}} = \frac{1}{3}\left(\boldsymbol{H}_{\text{h}} + \boldsymbol{H}_{\text{v}} + \boldsymbol{H}_{\text{d}}\right). \tag{14}$$

The bridge then computes a reliability map $\boldsymbol{\rho}_2$ following the formulation in (3). A gate $\mathcal{G}_{1\times1}(\cdot)$ (with a $5 \rightarrow 16 \rightarrow 3$ channel configuration, SiLU, and zero-initialized final weights/biases) predicts the mixing weights

$$\begin{aligned} \boldsymbol{\omega} = \text{softmax}\Big(\mathcal{G}_{1\times1}\big[&\eta\left(\boldsymbol{B}\right), \eta\left(\boldsymbol{L}\right), \eta\left(\boldsymbol{H}\right) \\ &\eta\left(\boldsymbol{H}_{\text{dir}}\right), \text{clip}\left(\boldsymbol{\rho}_2, 0, 8\right)\big]\Big) \end{aligned} \tag{15}$$

where $\boldsymbol{\omega} = \{\omega_{\text{l}}, \omega_{\text{h}}, \omega_{\text{dir}}\}$ are softmax weights. Before final aggregation, each component is independently refined: $\boldsymbol{L}' = \varphi_{1\times1}(\boldsymbol{L})$, $\boldsymbol{H}' = \varphi_{1\times1}(\boldsymbol{H})$, and $\boldsymbol{H}'_{\text{dir}} = \varphi_{1\times1}(\boldsymbol{H}_{\text{dir}})$. The WMB output is then formed as a residual-enhanced combination

$$\boldsymbol{X}_{\text{wmb}} = \boldsymbol{B} + \alpha_2\,\varphi_{1\times1}\left(\omega_{\text{l}}\boldsymbol{L}' + \omega_{\text{h}}\boldsymbol{H}' + \omega_{\text{dir}}\boldsymbol{H}'_{\text{dir}}\right) \tag{16}$$

where residual scale α_2 is initialized to 0.1.

E. Contour-Selective Fusion

The CSF performs contour-selective fusion at each PAN stage s, enhancing boundary-aware features before the OSC. For each stage $s \in \{2, 3, 4\}$, the input $\boldsymbol{U}_s$ is first processed by a C3 block to obtain a base feature

$$\boldsymbol{B}_s = \text{C3}\left(\boldsymbol{U}_s\right) \tag{17}$$

followed by the same low-/high-frequency decomposition used in WSB and WMB

$$\boldsymbol{L}_s = \mathcal{A}_{3\times3}\left(\boldsymbol{B}_s\right), \quad \boldsymbol{H}_s = \boldsymbol{B}_s - \boldsymbol{L}_s. \tag{18}$$

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Inspired by contourlet and steerable directional representations [52], [53], [54], CSF uses the same kernel geometries as in WSB

$$\begin{aligned} \boldsymbol{C}_s^{\mathrm{h}} &= \psi_{1\times 5}\left(\boldsymbol{H}_s\right), & \boldsymbol{C}_s^{\mathrm{v}} &= \psi_{5\times 1}\left(\boldsymbol{H}_s\right) \\ \boldsymbol{C}_s^{\mathrm{d}} &= \psi_{3\times 3}\left(\boldsymbol{H}_s\right), & \boldsymbol{C}_s^{\mathrm{c}} &= \psi_{3\times 3}^2\left(\boldsymbol{H}_s\right). \end{aligned} \tag{19}$$

A directional gate (with a $7 \to 16 \to 4$ channel configuration, SiLU, and zero-initialized final weights/biases) takes the channel-wise concatenation of seven response statistics to predict the directional fusion weights

$$\begin{aligned} \boldsymbol{\pi}_s = \operatorname{softmax}\Big(\mathcal{G}_{1\times 1}\big[&\eta\left(\boldsymbol{B}_s\right), \eta\left(\boldsymbol{L}_s\right), \eta\left(\boldsymbol{H}_s\right) \\ &\eta\left(\boldsymbol{C}_s^{\mathrm{h}}\right), \eta\left(\boldsymbol{C}_s^{\mathrm{v}}\right), \eta\left(\boldsymbol{C}_s^{\mathrm{d}}\right), \eta\left(\boldsymbol{C}_s^{\mathrm{c}}\right)\big]\Big) \end{aligned} \tag{20}$$

where $\boldsymbol{\pi}_s = \{\pi_s^{\mathrm{h}}, \pi_s^{\mathrm{v}}, \pi_s^{\mathrm{d}}, \pi_s^{\mathrm{c}}\}$ contains four softmax weights. The fused directional response is then obtained via weighted summation

$$\boldsymbol{C}_s = \sum_{i\in\{\mathrm{h,v,d,c}\}} \pi_s^i \boldsymbol{C}_s^i. \tag{21}$$

Before the final aggregation, the low-frequency, high-frequency, and contour components are independently refined: $\boldsymbol{L}_s' = \varphi_{3\times 3}(\boldsymbol{L}_s)$, $\boldsymbol{H}_s' = \varphi_{3\times 3}(\boldsymbol{H}_s)$, and $\boldsymbol{C}_s' = \varphi_{3\times 3}(\boldsymbol{C}_s)$. A second gate (with a $4 \to 16 \to 3$ channel configuration, SiLU, and zero-initialized final weights/biases) then predicts the mixing weights

$$\begin{aligned} \boldsymbol{\gamma}_s = \operatorname{softmax}\Big(\mathcal{G}_{1\times 1}\big[&\eta\left(\boldsymbol{B}_s\right), \eta\left(\boldsymbol{H}_s'\right) \\ &\eta\left(\boldsymbol{L}_s'\right), \eta\left(\boldsymbol{C}_s'\right)\big]\Big) \end{aligned} \tag{22}$$

where $\boldsymbol{\gamma}_s = \{\gamma_s^{\mathrm{h}}, \gamma_s^{\mathrm{l}}, \gamma_s^{\mathrm{c}}\}$ weights high, low, and contour candidates.

The final CSF output is formed via a residual aggregation

$$\boldsymbol{X}_{\mathrm{csf},s} = \boldsymbol{B}_s + \alpha_s \varphi_{3\times 3}\left(\gamma_s^{\mathrm{h}} \boldsymbol{H}_s' + \gamma_s^{\mathrm{l}} \boldsymbol{L}_s' + \gamma_s^{\mathrm{c}} \boldsymbol{C}_s'\right) \tag{23}$$

where α_s is initialized to 0.1. The three refinement mappings and the output projection are independent 3×3 Conv–BN–SiLU blocks.

F. Omnispectral Communication

OSC summarizes each local spectrum into magnitude-based radial-angular tokens and mixes them with the GRSS. The mixed tokens generate weights that modulate the retained complex spectra without changing their phase. Inverse transforms then reconstruct spatial residuals for gated injection.

The OSC operates on $\boldsymbol{X}_{\mathrm{csf},s} \in \mathbb{R}^{C_s\times H_s\times W_s}$ for each stage $s \in \{2, 3, 4\}$. A channel projection $\mathcal{P}_{\mathrm{chn}}(\cdot)$ is applied only when the input and output channel numbers differ, and stride-1 average pooling provides anti-aliasing smoothing (with padding 1, excluding padded samples from the averaging operation)

$$\boldsymbol{X}_{0,s} = \mathcal{P}_{\mathrm{chn}}\left(\boldsymbol{X}_{\mathrm{csf},s}\right), \quad \boldsymbol{X}_{\mathrm{aa},s} = \mathcal{A}_{3\times 3}\left(\boldsymbol{X}_{0,s}\right) \tag{24}$$

where $\boldsymbol{X}_{0,s}$ is the channel-aligned feature and $\boldsymbol{X}_{\mathrm{aa},s}$ is its anti-aliased version.

The feature $\boldsymbol{X}_{\mathrm{aa},s}$ is zero-padded on the right and bottom to multiples of $w = 8$ as needed, and then partitioned into Q nonoverlapping $w \times w$ windows $\{\boldsymbol{W}_q\}_{q=1}^{Q}$. Instead of full-map FFT, this local-window partitioning preserves region-specific spectral descriptions and limits the mixing of weak small-object responses with distant structured clutter. For each local-window index $q \in \{1, \ldots, Q\}$ and feature-channel index $c \in \{1, \ldots, C_s\}$, a one-sided complex spectrum is computed as

$$\boldsymbol{F}_{q,c} = \mathrm{rFFT}_2\left(\boldsymbol{W}_{q,c}\right) \tag{25}$$

where $\boldsymbol{W}_{q,c} \in \mathbb{R}^{w\times w}$ is the cth feature channel of window q; $\boldsymbol{F}_{q,c} \in \mathbb{C}^{w\times(w/2+1)}$ is its one-sided complex spectrum; and $\mathrm{rFFT}_2(\cdot)$ denotes the orthonormally scaled 2-D real-input FFT [63], [64], [65]. These window-local spectra preserve 2-D neighborhood support and avoid serializing an entire high-resolution feature map into one long sequence.

1) Radial-Angular Spectral Tokens: Let u and v denote the horizontal and vertical frequency coordinates in cycles per pixel, sampled by rfftfreq(w) and fftfreq(w) with unit sample spacing, respectively. Their Cartesian product forms the one-sided frequency grid Ω_f. The normalized radius and orientation angle are

$$r(u, v) = \frac{\sqrt{u^2 + v^2}}{\max\left(\max_{(u',v')\in\Omega_f} \sqrt{u'^2 + v'^2}, \varepsilon\right)} \tag{26}$$

$$\times\ \theta(u, v) = \operatorname{atan2}(v, u + \varepsilon) \bmod \pi \tag{27}$$

where $\varepsilon = 10^{-6}$. The indices $b \in \{1, \ldots, B\}$ and $a \in \{1, \ldots, A\}$ identify radial bands and angular bins, respectively. We partition the normalized radius into B bands using exponent $\kappa = 0.6$

$$\mathcal{R}_b = \begin{cases} [e_{b-1}, e_b), & b < B \\ [e_{B-1}, e_B], & b = B \end{cases}, \quad e_b = \left(\frac{b}{B}\right)^{\kappa} \tag{28}$$

and the orientation is uniformly quantized into A bins

$$\Theta_a = \begin{cases} [(a-1)\pi/A, a\pi/A), & a < A \\ [(A-1)\pi/A, \pi], & a = A. \end{cases} \tag{29}$$

The radial-angular mask is defined as

$$\boldsymbol{M}_{b,a}(u, v) = \mathbb{I}\left[r(u, v) \in \mathcal{R}_b\right] \mathbb{I}\left[\theta(u, v) \in \Theta_a\right] \tag{30}$$

where $\mathbb{I}$ is the indicator function; the mask is one inside the intersection of radial band $\mathcal{R}_b$ and angular bin Θ_a, and zero otherwise.

Let $F_{q,c}(u, v)$ denote the complex spectral coefficient at $(u, v) \in \Omega_f$. For window q, each radial-angular pair (b, a) yields a C_s-dimensional token whose cth feature-channel component is

$$t_{q,b,a,c} = \frac{\sum_{(u,v)\in\Omega_f} \boldsymbol{M}_{b,a}(u, v) \max\left(|F_{q,c}(u, v)|, \varepsilon\right)}{\max\left(\sum_{(u,v)\in\Omega_f} \boldsymbol{M}_{b,a}(u, v), 1\right)}. \tag{31}$$

Ordering the (b, a) pairs yields the per-window token sequence $\boldsymbol{T}_q \in \mathbb{R}^{BA\times C_s}$. Stacking the sequences from all Q windows gives $\boldsymbol{T} = \{\boldsymbol{T}_q\}_{q=1}^{Q} \in \mathbb{R}^{Q\times BA\times C_s}$. The GRSS processes each $\boldsymbol{T}_q$ independently, without propagating the state across windows. For WSCOM-YOLO-Lite, we set $A = 1$ and $\Theta_1 = [0, \pi]$. Consequently, the radial-angular tokenization reduces to radial-only tokenization to yield the lite sequence, while the AdaBand, GRSS, reconstruction, and spatial injection remain unchanged.

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TABLE X

THREE-SEED REPEATABILITY RESULTS AND CAPACITY-MATCHED CONTROL ON THE VISDRONE VALIDATION SET USING THE FIXED-VERSION ULTRALYTICS EVALUATOR (MEAN ± SAMPLE STANDARD DEVIATION, %)

Variant	Precision	Recall	mAP50	mAP50-95
M0 (YOLOv5s+P2)	57.54 ± 1.27	44.99 ± 0.59	45.63 ± 0.20	27.30 ± 0.09
M1 (no-WSB)	59.44 ± 1.40	47.75 ± 0.16	48.43 ± 0.45	29.43 ± 0.32
M2 (no-WMB)	59.49 ± 0.55	47.32 ± 0.18	48.34 ± 0.09	29.41 ± 0.13
M4 (no-CSF)	57.63 ± 0.79	45.86 ± 0.75	46.08 ± 0.35	27.61 ± 0.24
M5 (no-OSC)	58.71 ± 0.55	47.26 ± 0.25	47.69 ± 0.19	28.84 ± 0.03
M7 (WSCOM-YOLO)	58.84 ± 0.95	$\mathbf{48.15 \pm 0.38}$	$\mathbf{48.59 \pm 0.23}$	29.41 ± 0.23
GFLOPs-matched M0	57.49 ± 1.23	45.30 ± 0.54	45.62 ± 0.12	27.28 ± 0.09
WSCOM-YOLO-Lite	$\mathbf{59.68 \pm 0.77}$	47.69 ± 0.20	48.55 ± 0.37	$\mathbf{29.50 \pm 0.20}$

TABLE XI

SEED-0 DITHER AND CLEAN CSF BRANCH-REMOVAL ABLATIONS ON THE VISDRONE VALIDATION SET USING THE FIXED-VERSION ULTRALYTICS EVALUATOR (%)

Variant	mAP50	mAP50-95
Dither off	48.47	29.36
clean CSF (no-H)	48.06	29.24
clean CSF (no-V)	48.62	29.51
clean CSF (no-D)	48.20	29.34
clean CSF (no-DC)	48.04	29.10
M7 (seed 0)	48.84	29.66

Note: H/V/D/DC: horizontal, vertical, local/diagonal, and dilated-contour branches. Seed 0; maximum 300 epochs; patience 100. Completed epochs (best epoch): no-H 300 (282), no-V 297 (197), no-D 300 (207), no-DC 287 (187).

2) AdaBand Amplification and Spectral-Token Dither: AdaBand performs channel-wise reweighting on each radial-angular token via image-adaptive gains $g_{b,c}$, which are indexed by the radial band b and feature channel c. For each input image, identical gains are applied to channel c for all tokens belonging to band b, covering all angular bins and local windows. This weight-sharing strategy does not aggregate tokens, thereby preserving their angular and window distinctions. The gain is derived via global average pooling (GAP) followed by two pointwise convolutions, formulated as

$$g_{b,c} = 1 + \lambda_{\mathrm{a}}\, \sigma\left(\mathcal{I}_{b,c}\left(\phi\left(\mathrm{GAP}\left(\boldsymbol{X}_{0,s}\right)\right)\right)\right) \tag{32}$$

and the tokens are reweighted as

$$\widetilde{t}_{q,b,a,c} = g_{b,c}\, t_{q,b,a,c} \tag{33}$$

where $\phi(\cdot)$ is a biased 1×1 Conv–ReLU–Conv block (with a $C_s \to \max(\lfloor C_s/4 \rfloor, 1) \to BC_s$ channel configuration); $\lambda_{\mathrm{a}} = 0.5$ bounds the additive gain; $\sigma(\cdot)$ denotes the sigmoid function; $\mathcal{I}_{b,c}(\cdot)$ reshapes the BC_s-channel output into a $B \times C_s$ array and selects its (b, c) entry; and GAP$(\cdot)$ denotes the global average pooling.

During training, weak Gaussian perturbation is injected into the tail tokens whose indices belong to the set $\mathcal{K}_{\mathrm{d}}$

$$\widetilde{\boldsymbol{T}}_{q,k,:} \leftarrow \widetilde{\boldsymbol{T}}_{q,k,:} + \boldsymbol{\xi}_{q,k,:}, \quad \boldsymbol{\xi}_{q,k,:} \sim \mathcal{N}\left(\boldsymbol{0}, \sigma_{\mathrm{d}}^2 \boldsymbol{I}_{\mathrm{d}}\right) \tag{34}$$

where $k = (b-1)A + a \in \mathcal{K}_{\mathrm{d}}$ indexes the radial-angular sequence; N_{d} is the configured tail-token limit, and $\mathcal{K}_{\mathrm{d}}$ contains the final $\min(N_{\mathrm{d}}, BA)$ token indices, so $|\mathcal{K}_{\mathrm{d}}| = \min(N_{\mathrm{d}}, BA)$. The actual dithered-token counts at P2/P3/P4 are 8/8/0 for WSCOM-YOLO and 2/3/0 for WSCOM-YOLO-Lite. $\boldsymbol{I}_{\mathrm{d}}$ is the identity matrix; $\sigma_{\mathrm{d}} = 10^{-4}$ is the nonzero dither standard deviation; and the per-level settings are given in Table I. Dither introduces small stochastic perturbations to selected spectral tokens during training, providing token-level regularization; it is disabled during validation and inference. Its effect is examined by the seed-0 dither-off control in Section IV-D3.

3) Gated Recurrent Spectral Scan: Given the token sequence $\widetilde{\boldsymbol{T}}_q$, GRSS first projects each token into a content representation $\boldsymbol{V}_q$ and a set of gate logits $\boldsymbol{G}_q^{\mathrm{tok}}$ via a shared linear transformation, followed by a depthwise 1-D convolution $\mathcal{C}_{1-\mathrm{D}}^{\mathrm{dw}}(\cdot)$ that aggregates adjacent tokens

$$\left[\boldsymbol{V}_q, \boldsymbol{G}_q^{\mathrm{tok}}\right] = \widetilde{\boldsymbol{T}}_q \boldsymbol{W}_{\mathrm{in}}^{\mathrm{T}} + \boldsymbol{1}_{BA} \boldsymbol{b}_{\mathrm{in}}^{\mathrm{T}}, \quad \boldsymbol{V}_q, \boldsymbol{G}_q^{\mathrm{tok}} \in \mathbb{R}^{BA \times C_s} \tag{35}$$

$$\boldsymbol{u}_{q,k} = \mathcal{I}_k\left(\mathcal{C}_{1-\mathrm{D}}^{\mathrm{dw}}\left(\boldsymbol{V}_q\right)\right) \tag{36}$$

where $\boldsymbol{W}_{\mathrm{in}} \in \mathbb{R}^{2C_s \times C_s}$ and $\boldsymbol{b}_{\mathrm{in}} \in \mathbb{R}^{2C_s}$ implement a $C_s \to 2C_s$ linear transformation; $\boldsymbol{1}_{BA}$ is an all-ones column vector; $\mathcal{I}_k(\cdot)$ extracts the kth element of a sequence; and $\mathcal{C}_{1-\mathrm{D}}^{\mathrm{dw}}(\cdot)$ uses kernel size 3, padding 1, C_s groups, and learnable bias terms. GRSS then propagates the recurrent state

$$\boldsymbol{s}_{q,k} = \tanh\left(\beta_{\mathrm{ret}} \boldsymbol{s}_{q,k-1} + \boldsymbol{W}_{\mathrm{s}} \boldsymbol{u}_{q,k} + \boldsymbol{b}_{\mathrm{s}}\right) \tag{37}$$

$$\boldsymbol{z}_{q,k} = \boldsymbol{W}_{\mathrm{p}}\left(\left(\boldsymbol{W}_{\mathrm{o}} \boldsymbol{s}_{q,k} + \boldsymbol{b}_{\mathrm{o}}\right) \odot \sigma\left(\boldsymbol{G}_{q,k}^{\mathrm{tok}}\right)\right) + \boldsymbol{b}_{\mathrm{p}} \tag{38}$$

where $\boldsymbol{s}_{q,k}$ is the recurrent state, initialized independently per window as $\boldsymbol{s}_{q,0} = \boldsymbol{0}$; $\beta_{\mathrm{ret}} = 0.7$ is the fixed state-retention coefficient shared by all levels and experiments; $\boldsymbol{W}_{\mathrm{s}}$ maps content features into the state space; $\boldsymbol{W}_{\mathrm{o}}$ maps the state to the gated feature space; $\boldsymbol{W}_{\mathrm{p}}$ produces the output token $\boldsymbol{z}_{q,k}$; and $\odot$ denotes the element-wise multiplication. The state dimension d_{state} is set to 16, with $\boldsymbol{W}_{\mathrm{s}} \in \mathbb{R}^{16 \times C_s}$, $\boldsymbol{W}_{\mathrm{o}} \in \mathbb{R}^{C_s \times 16}$, and $\boldsymbol{W}_{\mathrm{p}} \in \mathbb{R}^{C_s \times C_s}$, each having a corresponding output-sized bias term $\boldsymbol{b}_{\mathrm{s}}$, $\boldsymbol{b}_{\mathrm{o}}$, and $\boldsymbol{b}_{\mathrm{p}}$, respectively. This gated recurrence implements a short-sequence mixer [78], [79].

4) Spectral Reconstruction and Spatially Gated Injection: The mixed tokens generate a real-valued spectral gate $\boldsymbol{G}_{q,c}^{\mathrm{spec}}$ over the one-sided spectrum

$$\boldsymbol{G}_{q,c}^{\mathrm{spec}}(u, v) = \sum_{b=1}^{B} \sum_{a=1}^{A} \sigma\left(z_{q,b,a,c}\right) \boldsymbol{M}_{b,a}(u, v) \tag{39}$$

and each spatial window is reconstructed by

$$\boldsymbol{S}_{q,c} = \mathrm{irFFT}_2\left(\boldsymbol{F}_{q,c} \odot \boldsymbol{G}_{q,c}^{\mathrm{spec}}\right) \tag{40}$$

where $z_{q,b,a,c}$ is the scalar component of the mixed token associated with window q, radial-angular pair (b, a), and channel c; $\mathrm{irFFT}_2(\cdot)$ is the inverse real-input FFT; and $\boldsymbol{S}_{q,c}$ is the reconstructed spatial residual window. The windows are then reassembled, and the padding is removed, yielding $\boldsymbol{S}$. A 3×3 Conv–BN–SiLU block subsequently produces the residual tensor $\boldsymbol{R}$

$$\boldsymbol{R} = \varphi_{3\times 3}(\boldsymbol{S}). \tag{41}$$

After restoring the scale index to obtain $\boldsymbol{R}_s$, we construct the spatial gate as $\boldsymbol{G}_s^{\mathrm{spa}} = \sigma(\mathcal{C}_{1\times 1}(\boldsymbol{X}_{0,s}))$. The OSC output delivered to the sth detection branch is

$$\boldsymbol{P}_s = \boldsymbol{X}_{0,s} + \alpha_{\mathrm{o}}\, \boldsymbol{G}_s^{\mathrm{spa}} \odot \boldsymbol{R}_s \tag{42}$$

where $\mathcal{C}_{1\times 1}(\cdot)$ is a zero-initialized linear convolution without batch normalization or activation; and the learnable residual

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scale α_o is initialized to 0.25. The gate provides a learned spatial modulation of the reconstructed spatial residual.

OSC combines local radial-angular token communication with spectral residual reconstruction and spatially gated injection. The spectral gate modulates frequency bins during reconstruction, whereas the spatial gate controls where the reconstructed residual is injected.

G. Detection Loss and Training Objective

All modules are trained end-to-end under the detection objective. Task-aligned assignment [80] with top-$k = 10$, $\alpha = 0.5$, and $\beta = 6.0$ supplies soft targets. The Detect head uses reg_max = 16 (distance bins 0–15); distribution focal loss (DFL) targets are clamped to $[0, 14.99]$ before interpolation. The detector is optimized with complete IoU (CIoU) loss, binary cross-entropy (BCE), and DFL [81]

$$\mathcal{L} = \lambda_{\text{box}}\mathcal{L}_{\text{CIoU}} + \lambda_{\text{cls}}\mathcal{L}_{\text{BCE}} + \lambda_{\text{dfl}}\mathcal{L}_{\text{DFL}} \tag{43}$$

where the λ terms are scalar gains. Let Ω denote all locations, Ω_+ the assigned foreground, and $\boldsymbol{y}_i = (y_{i,1}, \ldots, y_{i,K})$ the soft class target ($\mathbf{0}$ for background). Define $w_i = \sum_k y_{i,k}$ and $\mathcal{Z} = \max\left(\sum_i w_i, 1\right)$. The CIoU loss is

$$\mathcal{L}_{\text{CIoU}} = \frac{1}{\mathcal{Z}} \sum_{i\in\Omega_+} w_i \left[1 - \text{CIoU}\left(\widehat{\boldsymbol{b}}_i, \boldsymbol{b}_i\right)\right] \tag{44}$$

where $\widehat{\boldsymbol{b}}_i$ and $\boldsymbol{b}_i$ are predicted and assigned boxes. The classification loss is

$$\mathcal{L}_{\text{BCE}} = \frac{1}{\mathcal{Z}} \sum_{i\in\Omega} \sum_{k=1}^{K} \text{BCE}\left(\ell_{i,k}^{\text{cls}}, y_{i,k}\right) \tag{45}$$

where $\ell_{i,k}^{\text{cls}}$ is the class logit and $\text{BCE}(\ell, y) = -y \log \sigma(\ell) - (1 - y) \log[1 - \sigma(\ell)]$. For box side $e \in \{\text{l}, \text{t}, \text{r}, \text{b}\}$, let target distance $d_{i,e}$ have adjacent bins $d_{i,e}^- = \lfloor d_{i,e} \rfloor$ and $d_{i,e}^+ = d_{i,e}^- + 1$. The DFL loss is

$$\ell_{\text{DFL}}\left(\boldsymbol{\zeta}, d\right) = \left(d^+ - d\right) \text{CE}\left(\boldsymbol{\zeta}, d^-\right) + \left(d - d^-\right) \text{CE}\left(\boldsymbol{\zeta}, d^+\right)$$

$$\mathcal{L}_{\text{DFL}} = \frac{1}{4\mathcal{Z}} \sum_{i\in\Omega_+} w_i \sum_{e\in\{\text{l,t,r,b}\}} \ell_{\text{DFL}}\left(\boldsymbol{\zeta}_{i,e}, d_{i,e}\right) \tag{46}$$

where $\boldsymbol{\zeta}_{i,e}$ are distance-bin logits; $\text{CE}(\cdot, \cdot)$ is cross-entropy; and $1/4$ averages box sides. Ultralytics gains are $\lambda_{\text{box}} = 7.5$, $\lambda_{\text{cls}} = 0.5$, and $\lambda_{\text{dfl}} = 1.5$. The common batch-size multiplier is omitted because it is a constant factor, and the objective contains only the box, classification, and DFL terms.

IV. Experiments

A. Experimental Settings

1) Datasets: VisDrone2019-DET is the primary benchmark, while DIOR and AI-TODv2 are used to evaluate higher-altitude and tiny-object settings.

VisDrone2019-DET: VisDrone2019-DET [1] is a UAV-based object detection benchmark containing 10 209 images: 6471 for training, 548 for validation, and 3190 for testing (1610 test-dev and 1580 test-challenge). It covers ten categories, including pedestrians and vehicles, in UAV images with dense objects, frequent occlusion, and complex backgrounds. We use the official training split for training and the validation split for model selection and ablation studies. Quantitative testing uses the 1610 test-dev images with publicly available annotations. The test-challenge split is excluded from local quantitative evaluation because its annotations are not publicly available.

DIOR: DIOR [17] contains 23 463 800 × 800 optical remote-sensing images, 192 472 instances, and 20 categories, split into 5862/5863/11 738 training/validation/test images. Its greater land-cover diversity and scale variation extend evaluation beyond low-altitude traffic scenes.

AI-TODv2: AI-TODv2 [19] has 11 214/2804/14 018 training/validation/test images and 301 534/75 091/376 121 annotations over eight classes: airplane, bridge, storage tank, ship, swimming pool, vehicle, person, and windmill. It emphasizes tiny, localization-sensitive objects and complements the other two benchmarks. COCO annotations are converted into the YOLO format without changing dataset splits, category identities, or bounding-box geometries.

2) Implementation Settings: We implement WSCOM-YOLO in Ultralytics and train it on a single RTX 4090. Our baseline, denoted as M0, is a YOLOv5s+P2 adaptation within the same framework. It adopts a YOLOv5s-style C3/SPPF backbone-neck architecture, P2–P5 branches, and an Ultralytics anchor-free detection head. M1–M7 correspond to the IDs of our ablation configurations (Section IV-D1). Specifically, M4, M5, and M7 represent the no-CSF variant, the no-OSC variant, and WSCOM-YOLO, respectively.

YOLOv5s+P2 and WSCOM-YOLO use depth/width multipliers of 0.33/0.50. In YOLOv5s+P2, standard C3 and convolutional blocks replace WSB, WMB, CSF, and OSC; the detection head, losses, sample assignment, and training protocol are unchanged. This baseline is not the official anchor-based YOLOv5s.

Unless otherwise stated, the same-framework models are trained from scratch at 640 × 640 for up to 300 epochs. Optimization uses stochastic gradient descent (SGD) with an initial learning rate of 0.01, a final ratio of 0.01, momentum 0.937, weight decay 5×10^{-4}, a linear schedule with three warmup epochs, and automatic mixed precision (AMP). The physical training batch size is 8 with a nominal batch size of 64 (`nbs = 64`), corresponding to approximately eight accumulated batches per update after warmup. To ensure a fair and reliable evaluation, M0, M1, M2, M4, M5, M7, WSCOM-YOLO-Lite, billions of floating-point operations (GFLOPs)-matched M0, and YOLO11s+P2 use seeds 0–2 on the VisDrone validation set; the remaining controlled ablations use seed 0 unless otherwise specified.

The fixed-version Ultralytics 8.4.41 evaluator combined with PyTorch 2.1.2+cu118 adopts 32-bit floating-point (FP32) inference and an evaluation batch size of 8. Images are resized with preserved aspect ratio so that the longer side is 640 pixels, followed by stride-aligned rectangular padding within each batch. Therefore, 640 denotes the nominal resize setting, rather than a 640 × 640 padded tensor; the bounding boxes undergo the corresponding coordinate transforms.

Evaluation employs a confidence floor of 0.001 without test-time augmentation. Class-aware nonmaximum suppression

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(NMS) uses an IoU threshold of 0.7 and retains at most 300 detections per image. The confidence floor preserves predictions for precision-recall construction. Reported precision and recall are read at the confidence index that maximizes the smoothed mean $F1$ curve and are then macro-averaged across classes. Checkpoints are selected based on validation fitness during training and frozen prior to test evaluation. Unlike this batched accuracy evaluation, forward-latency profiling uses a fixed 640 × 640 input and batch size one.

3) Evaluation Metrics: For a predicted box $\widehat{\boldsymbol{b}}$ and ground-truth box $\boldsymbol{b}$, let $|\cdot|$ denote image-plane area. Their IoU is

$$\mathrm{IoU}\left(\widehat{\boldsymbol{b}}, \boldsymbol{b}\right) = \frac{|\widehat{\boldsymbol{b}} \cap \boldsymbol{b}|}{|\widehat{\boldsymbol{b}} \cup \boldsymbol{b}|}. \tag{47}$$

For each class, let TP, FP, and FN denote the numbers of true positives, false positives, and false negatives at the selected confidence operating point, with class-aware one-to-one matching at IoU ≥ 0.50. Then

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}, \qquad \mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}. \tag{48}$$

The reported precision and recall are the arithmetic means of these per-class values.

For class index $k \in \{1, \ldots, K\}$ and IoU threshold τ, $\mathrm{AP}_{k,\tau}$ is the area under the interpolated precision–recall curve of class k, where K is the number of evaluated classes. The reported mean average precisions (mAPs) are

$$\mathrm{mAP50} = \frac{1}{K} \sum_{k=1}^{K} \mathrm{AP}_{k,0.50} \tag{49}$$

$$\mathrm{mAP50\text{-}95} = \frac{1}{10K} \sum_{j=0}^{9} \sum_{k=1}^{K} \mathrm{AP}_{k,0.50+0.05j} \tag{50}$$

where $j \in \{0, \ldots, 9\}$ enumerates the ten IoU thresholds from 0.50 to 0.95 in increments of 0.05. We profile computational complexity using THOP (PyTorch-OpCounter), a tool for profiling PyTorch models, with batch size one and a 640 × 640 input. Its multiply-accumulate count a_{thop} is converted to GFLOPs, counting one multiplication and one addition per MAC

$$\mathrm{GFLOPs} = 2 \times 10^{-9}\, a_{\mathrm{thop}}. \tag{51}$$

The $\mathrm{rFFT}_2(\cdot)$, $\mathrm{irFFT}_2(\cdot)$, and other unsupported functional operators are excluded. Therefore, the reported GFLOPs are profiler-based estimates and are presented together with measured latency in the accuracy–complexity analysis.

The scale-specific analysis uses common pycocotools COCOeval rather than the fixed-version Ultralytics evaluator. Small, medium, and large object definitions follow the default COCO area ranges $[0, 32^2]$, $[32^2, 96^2]$, and $[96^2, 10^{10}]$ pixels^2, respectively, with area computed as bounding-box width times height in original-image pixel coordinates, not resized input coordinates. COCOeval retains its inclusive endpoint processing. Parameter count includes trainable and nontrainable model parameters.

B. Comparison With Related Methods

We first compare detectors on the VisDrone validation set, then report dataset-specific test results, and finally compare external detectors using common COCOeval. These comparisons assess complete detectors; controlled architectural contributions are examined separately in the ablation studies.

1) VisDrone Validation-Set Comparison: Table II compares detectors on the VisDrone validation set. Entries marked ‡ correspond to research-group experiments under method-specific settings; unmarked entries are evaluated via the fixed-version Ultralytics evaluator. The former offers broader empirical context rather than controlled architectural evidence.

Under the same fixed-version evaluator, WSCOM-YOLO achieves three-seed mean accuracies of 48.59% ± 0.23% mAP50 and 29.41% ± 0.23% mAP50–95, compared with 41.90% ± 0.19% and 24.94% ± 0.12% for YOLO11s+P2. The mean gains are 6.69 and 4.47 percentage points, respectively. WSCOM-YOLO also achieves higher mAP50 and mAP50–95 values than the method-specific entries in Table II, although their training and evaluation settings differ. The three-seed comparison of YOLO11s+P2 is conducted only on the VisDrone validation set; results on the VisDrone test set, DIOR, and AI-TODv2 remain based on single checkpoints.

2) Test-Set Comparisons Across Datasets: All test comparisons adopt the fixed-version Ultralytics evaluator after dataset-specific training, rather than direct cross-dataset transfer. The † marker identifies an Ultralytics adaptation with an anchor-free head, DFL, and task-aligned assignment. HIC-YOLOv5† is included across all three test-set comparisons.

YOLOv5s+P2† uses a historical single M0 checkpoint on VisDrone; the DIOR and AI-TODv2 adaptations differ in configuration and are not controlled M0 ablations.

Table III reports results on the 1610-image VisDrone test-dev split. On this VisDrone test-dev split, WSCOM-YOLO improves mAP50/mAP50–95 by 2.07/1.50 percentage points over YOLOv5s+P2† and 2.35/1.75 points over HIC-YOLOv5†.

Table IV presents DIOR test results. On DIOR, WSCOM-YOLO yields mAP50/mAP50–95 gains of 0.46/0.93 percentage points over YOLO11s+P2 and 1.33/2.71 points over HIC-YOLOv5†.

Table V extends the test comparison to AI-TODv2, with a focus on tiny-object detection. On AI-TODv2, WSCOM-YOLO outperforms HIC-YOLOv5† by 1.32 mAP50–95 percentage points.

3) Common-COCOeval Comparison: Table VI compares external detectors on the VisDrone validation set using common pycocotools COCOeval, shared ground truth, a ten-category mapping, a nominal input size of 640, and an identical detection cap. Note that HIC-YOLOv5 in this table refers to the authors' anchor-based implementation, rather than our local Ultralytics-adapted variant.

All entries in Table VI are evaluated using one fixed checkpoint with the same COCOeval protocol. This comparison does not measure variation across training runs. Because training and postprocessing settings differ between methods, the observed differences cannot be attributed solely to model architecture. Repeatability of the main comparisons is assessed

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separately by the three-seed results reported in Table II and Section IV-D2.

DCP-Net is trained from scratch using SGD for 300 epochs; SMWG-DETR adopts its official ImageNet-pretrained ResNet-50 initialization and 36-epoch training recipe; and HIC-YOLOv5 employs COCO-pretrained weights and Adam for 300 epochs. Initialization, optimization, gradient accumulation, and augmentation therefore remain method-specific.

Under common COCOeval, WSCOM-YOLO exceeds DCP-Net by 9.33/6.56, SMWG-DETR by 20.37/14.83, and HIC-YOLOv5 by 8.26/6.00 mAP50/mAP50–95 percentage points. HIC-YOLOv5 and WSCOM-YOLO use NMS IoU thresholds of 0.6 and 0.7, respectively. The shared evaluator aligns metric computation, but not all training or postprocessing configurations.

Using the same seed-0 checkpoint of WSCOM-YOLO, the fixed-version Ultralytics evaluator yields 48.84/29.66 (Section IV-D3), whereas common COCOeval yields 48.91/29.82. The differences of 0.07/0.16 points demonstrate why results obtained from these two evaluators must be reported separately.

C. Accuracy-Complexity Analysis

Same-framework complexity and forward inference speed are shown in Table VII. Relative to YOLOv5s+P2, WSCOM-YOLO adds 87.44 GFLOPs and 18.289 $\pm$ 0.060 ms of forward latency, with three-seed mean gains of 2.96/2.11 points in mAP50/mAP50–95.

The GFLOPs-matched M0 control YOLOv5s+P2 retains a depth multiplier of 0.33, uses a width multiplier of 0.625, and increases the P2's C3 block to 18 internal bottleneck repeats after depth scaling. The configuration was selected using only parameter count and profiled GFLOPs, without referring to validation or test accuracy.

WSCOM-YOLO-Lite reduces token counts from 8/24/24 to 2/3/3 and forward latency by 41.78%. Its three-seed mean mAP50/mAP50–95 differs from WSCOM-YOLO by −0.04/+0.09 points on VisDrone. In the DIOR seed-0 pair, WSCOM-YOLO achieves 73.09%/52.55% mAP50/mAP50–95 versus 72.84%/52.16% for WSCOM-YOLO-Lite, a 0.25/0.39-point advantage. These results indicate a dataset-dependent accuracy-latency trade-off.

Table VIII shows that CSF contributes the largest profiled GFLOPs increment, whereas OSC has the largest measured latency impact. WSB and WMB introduce comparatively small overheads.

D. Ablation Study

The ablation study follows the processing pipeline of the model. The complete M0–M7 suite first evaluates the front-end WSB/WMB modules and the prehead CSF-OSC pathway. Three-seed experiments are then performed to assess repeatability and the capacity-matched comparison, followed by dither, branch, and object-scale analyses.

1) Single-Checkpoint Module Ablation: Table IX summarizes the M0–M7 module configurations, class-wise AP50, and overall mAP50/mAP50–95. WSCOM-YOLO outperforms YOLOv5s+P2 by 3.63/2.46 mAP points. Variant M3, which retains the CSF–OSC pathway, achieves performance comparable to M7, whereas variant M6, which retains only WSB/WMB, performs similarly to YOLOv5s+P2. This single-checkpoint observation indicates that WSB/WMB serve as auxiliary premodulation, while CSF/OSC contribute the majority of the performance gain. The formal three-seed estimates for the individual removals are presented in Table X, further supporting this observation. Per-class gains extend beyond the frequent car category, especially to bicycle, truck, tricycle, awning-tricycle, bus, and motor.

The M7 checkpoint in Table IX gives 49.04/29.67 in mAP50/mAP50–95, compared with 48.84/29.66 for the seed-0 run (Table XI) and 48.59 $\pm$ 0.23/29.41 $\pm$ 0.23 across three seeds (Table X). The main accuracy estimates use the three-seed mean, and the seed-0 controls use the seed-0 reference.

2) Three-Seed Repeatability: Table X presents three-seed repeatability results and capacity-matched control on the VisDrone validation set. Averaged over three seeds, WSCOM-YOLO achieves gains of 2.96/2.11 percentage points in mAP50/mAP50–95 over YOLOv5s+P2. The GFLOPs-matched M0 obtains 45.62 $\pm$ 0.12 mAP50 and 27.28 $\pm$ 0.09 mAP50–95. WSCOM-YOLO outperforms GFLOPs-matched M0 by 2.97/2.13 points despite their nearly matched nominal parameter counts and profiled GFLOPs. These results indicate that scaling the baseline to a comparable parameter count and profiled GFLOPs cannot replicate the accuracy improvements achieved by WSCOM-YOLO.

Compared with WSCOM-YOLO, M1 (no-WSB) differs by −0.16/+0.02 points and M2 (no-WMB) by −0.25/0.00 points in mAP50/mAP50–95. Such small mean differences do not support stable independent mAP50–95 gains from WSB or WMB; hence, we retain them as auxiliary front-end premodulation modules. Removing CSF reduces mAP50 and mAP50–95 by 2.51 and 1.80 percentage points, respectively. Removing OSC reduces them by 0.90 and 0.57 percentage points. These findings verify that CSF and OSC are the main contributors to WSCOM-YOLO's performance gains over the baseline.

WSCOM-YOLO-Lite achieves 48.55 $\pm$ 0.37 mAP50 and 29.50 $\pm$ 0.20 mAP50–95, differing from WSCOM-YOLO by −0.04/+ 0.09 points, with 41.78% lower forward latency (Table VII). Its VisDrone performance is therefore comparable to WSCOM-YOLO at much lower latency, confirming radial-only tokenization as an efficient design choice.

To further dissect the class-wise performance, training dynamics, and error patterns revealed by the multiseed ablation results, we provide complementary visualizations in Figs. 2–4.

Fig. 2 shows that WSCOM-YOLO improves the three-seed mean AP50 across multiple categories, indicating that the aggregate gain is not driven solely by the car category. The smaller and less uniform class-wise changes for M1 (no-WSB) and M2 (no-WMB) are consistent with their auxiliary role, whereas the substantial overall performance drops for M4 (no-CSF) and M5 (no-OSC) confirm that these two modules act as the main performance contributors.

Fig. 3 shows smoothly decreasing losses and stable late-stage validation trajectories, and the trajectories of M0, M4, M5, and M7 near convergence are consistent with the final means in Table X.

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TABLE XII
SCALE-SPECIFIC AP OF YOLOV5S+P2 AND WSCOM-YOLO USING SINGLE CHECKPOINTS ON THE VISDRONE VALIDATION SET (COMMON COCOEVAL; maxDets = 300; %)

Method	$AP_{s,50}$	$AP_{s,50:95}$	$AP_{m,50}$	$AP_{m,50:95}$	$AP_{l,50}$	$AP_{l,50:95}$
YOLOv5s+P2	36.43	18.73	58.82	38.77	58.46	43.69
WSCOM-YOLO	**40.15**	**20.67**	**62.01**	**41.26**	**63.59**	**47.67**

Fig. 4 exposes class-specific background confusion beyond the aggregated AP metrics. Under the shared display threshold, the WSCOM matrix has a stronger diagonal than YOLOv5s+P2 for pedestrian, people, bicycle, van, truck, tricycle, awning-tricycle, bus, and motor, while the car diagonal changes only slightly. The corresponding background row is also lower for most classes, matching the class-wise AP improvements. Because the matrices depend on one operating threshold and normalize each ground-truth column, they identify error structure alongside the threshold-integrated AP metrics.

3) Capacity, Dither, and CSF-Branch Ablations: Table XI presents the seed-0 dither and clean CSF branch-removal ablations. Disabling dither reduces mAP50/mAP50–95 by 0.37/0.30 points. The clean-removal strategy deletes the selected operator at every CSF location: the corresponding response and statistics are removed; the gate dimension sequence shrinks from $7 \rightarrow 16 \rightarrow 4$ to $6 \rightarrow 16 \rightarrow 3$; only three remaining responses are fused; and the shared contour-refinement layer and the final output projection remain unchanged. Clean removal slightly reduces parameters and GFLOPs and is therefore not capacity-preserving.

Relative to M7, clean no-H/no-V/no-D/no-DC reduce mAP50/mAP50–95 by 0.78/0.42, 0.22/0.15, 0.64/0.32, and 0.80/0.56 points, respectively, and both AP metrics decrease in each run. These are single-seed diagnostic experiments.

Combined with the three-seed capacity-matched comparison, these seed-0 controls assess nominal capacity, dither, and individual CSF branches, complementing the aggregate M0–M7 ablation suite.

Because the proposed pathway is applied mainly to the higher-resolution pyramid branches, we also investigate whether the overall performance gain persists across different object-size groups, rather than relying only on overall AP.

4) Scale-Specific Analysis: Table XII compares scale-specific AP on the 548-image VisDrone validation split using the same ground truth, category mapping, and common COCOeval with maxDets = 300. Table XII uses the historical single-checkpoint M0 and M7 predictions associated with the complete ablation suite; these checkpoints differ from the formal seed-0 checkpoint reported in Table VI. The overall AP50/AP50-95 values for YOLOv5s+P2 and WSCOM-YOLO are 45.52%/27.40% and 49.16%/29.87%, respectively.

The validation set contains 26 586 small, 11 105 medium, and 1068 large instances. WSCOM-YOLO improves small-object AP50/AP50-95 by 3.72/1.94 and medium-object AP50/AP50-95 by 3.19/2.49 percentage points over YOLOv5s+P2.

The scale-specific results align with the design rationale of the proposed pathway: CSF and OSC operate on the P2–P4 pyramid levels, where the stride-4 and stride-8 branches retain the spatial support most relevant to small and medium objects, while the P5 level retains its native spatial branch. The small-object gain is therefore the primary scale-level evidence for the design. The large-object subset is more than an order of magnitude smaller than the small-object subset, yet M7 attains higher $AP_{l,50}$ than YOLOv5s+P2.

E. Visualization Analysis

The examples from the three datasets illustrate localization behavior, residual errors, and intermediate feature transformations.

1) Detection Comparisons: Figs. 5–7 maintain a consistent comparison order across datasets: input image, YOLOv5s+P2, YOLO11s+P2, WSCOM-YOLO, and ground truth. All visualizations adopt dataset-specific seed-0 checkpoints tailored to each dataset, with a unified confidence threshold of 0.25, an NMS IoU threshold of 0.7, a nominal input size of 640, and maxDets = 300 for single-image inference. Note that the DIOR visual results are plotted on the validation set, while its quantitative evaluation is performed on the test set.

Fig. 5 uses dense low-altitude traffic scenes with zoomed crops for examining small objects and crowded boundaries. Fig. 6 shows harbor and storage-tank scenes, while Fig. 7 shows tiny airplanes and clustered vehicles. The cropped vehicle region from WSCOM-YOLO in Fig. 7(b) shows fewer misses but more false positives than the displayed baselines. These qualitative samples are selected for visualization; quantitative table metrics are used for comprehensive performance comparison.

2) Failure Cases: Fig. 8 further investigates two representative failure modes: missed or poorly localized objects and false positives in structured backgrounds. YOLOv5s+P2, YOLO11s+P2, and WSCOM-YOLO adopt a confidence threshold of ≥ 0.25 together with class-aware one-to-one matching at IoU ≥ 0.5. The reported counts correspond only to the visualized regions, rather than the full validation set.

In Fig. 8(a), WSCOM-YOLO has 39 true positives and 17 misses, compared with 42/14 for YOLOv5s+P2 and 39/17 for YOLO11s+P2. The purple callout marks a class-correct localization miss (IoU $= 0.435$). This case illustrates missed detections and localization errors under occlusion and scale variation. In Fig. 8(b), the false-positive counts are 17/11/8 for WSCOM-YOLO/YOLOv5s+P2/YOLO11s+P2, while the corresponding true-positive counts are 32/30/31. WSCOM-YOLO detects more objects in this crop but also produces more false positives, illustrating residual structured-background confusion.

3) Intermediate Feature Responses: Fig. 9 visualizes intermediate responses from the same VisDrone-trained WSCOM-YOLO model on VisDrone and DIOR inputs.

The WMB responses include objects and structured background edges. The CSF enhancement and the OSC effective injection [where the injection term refers to $\alpha_o \boldsymbol{G}^{spa} \odot \boldsymbol{R}$ in (42)] characterize feature transformations, and the OSC

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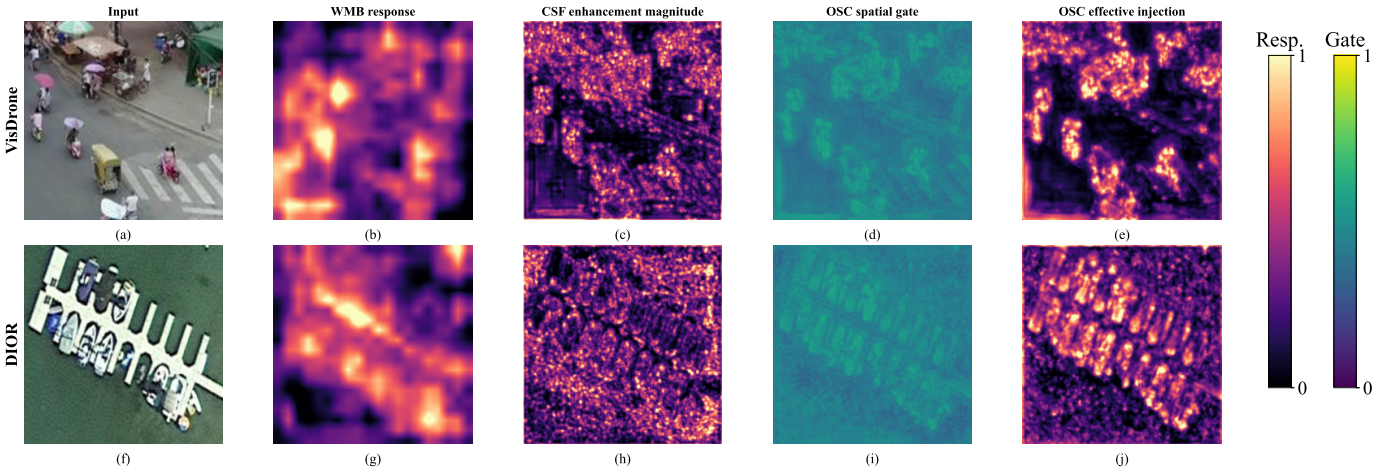


Fig. 9. VisDrone-trained M7 feature responses to VisDrone and DIOR inputs. Top row (VisDrone): (a) input image; (b) WMB response; (c) CSF enhancement magnitude; (d) OSC spatial gate; and (e) OSC effective injection. Bottom row (DIOR): (f) input image; (g) WMB response; (h) CSF enhancement magnitude; (i) OSC spatial gate; and (j) OSC effective injection. Channel-rms maps are independently clipped/normalized to their 1st–99th percentiles. Response intensities are not comparable across independently normalized maps or datasets. Gates retain $[0, 1]$; the maps show intermediate features.

gate applies spatial modulation over the range $[0, 1]$. Fig. 9 shows that WSCOM-YOLO produces nonuniform gates and effective injections on both VisDrone and DIOR scenes. Because response-map color scales are independently normalized, intensities are meaningful within each map rather than across maps or datasets. These maps visualize intermediate feature transformations instead of class-confidence scores. Accuracy and latency are quantified in the ablation and complexity analyses.

V. Conclusion

This article presents WSCOM-YOLO, in which WSB/WMB provide auxiliary frequency premodulation, and CSF-OSC performs directional conditioning, local spectral communication, and spatially gated residual injection. Three-seed comparisons on VisDrone2019-DET, including a GFLOPs-matched YOLOv5s+P2 control, provide evidence that the aggregate gains are repeatable and are not reproduced by this scaled standard-operator baseline, with larger differences associated with the prehead CSF-OSC pathway. Dataset-specific evaluations on DIOR and AI-TODv2 extend the analysis to additional remote-sensing settings. WSCOM-YOLO-Lite yields numerically similar three-seed means on VisDrone and 41.78% lower forward latency; the paired DIOR seed-0 test gives WSCOM-YOLO a modest 0.25/0.39-point mAP50/mAP50–95 advantage. Accordingly, the lite version is recommended when reducing forward latency is the primary objective. The full model should be considered when validation on the target dataset supports an accuracy benefit from finer angular tokenization and the additional forward latency is acceptable.

These results suggest that selectively conditioning local directional and spectral responses is a useful complement to preserving high-resolution features in aerial small-object detection. The comparison between WSCOM-YOLO and WSCOM-YOLO-Lite also shows that finer spectral partitioning must be weighed against measured latency and the accuracy requirements of the target dataset.

The current implementation has practical limitations: local FFT communication and spatial reconstruction add measurable forward latency, and the VisDrone test-set evaluation is limited to the 1610 labeled test-dev images.

The present evaluation is limited to RGB images, and its applicability to infrared and multispectral inputs has not been established. Differences in sensor characteristics, noise patterns, and image statistics may affect the learned directional and spatial-frequency responses. Extending the method to these modalities would require input-channel adaptation, modality-specific training, and dedicated evaluation.

Potential applications include traffic monitoring from aerial imagery and remote-sensing image analysis, with WSCOM-YOLO-Lite providing a lower-latency option. Future work will explore structured pruning and knowledge distillation, as well as adaptive token and band selection and fused local spectral processing, to reduce computational overhead and support deployment on resource-constrained devices.

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